A Sharp Maxwell Bound for Three Point Charges

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Abstract

Let $a_1, a_2, a_3 \in \mathbb{R}^3$ be distinct and let $q_i \in \mathbb{R} \setminus \{0\}$. We prove that the Coulomb potential $V_q(x) = \sum_{i=1}^{3} q_i/|x - a_i|$ has at most four distinct critical points whenever its critical set is finite. The bound is sharp. No nondegeneracy assumption is imposed, so isolated degenerate equilibria are included. The bound improves to two for collinear sites and to three for noncollinear sites with $\sum_i q_i = 0$. The proof combines a projective charge map, four elliptic discs, an exact Lami identity, degree theory, a cusp count, and an injectivity theorem for anti-periodic support envelopes.

1 Introduction

For point charges of strengths q_i at distinct sites $a_i \in \mathbb{R}^3$, the electrostatic potential is

$$V_q(x) = \sum_{i=1}^{n} \frac{q_i}{|x - a_i|}, \qquad x \in \mathbb{R}^3 \setminus \{a_1, \ldots, a_n\}.$$

In §113 of the first edition of his *Treatise*, Maxwell tracked changes in the periphraxy and cyclosis of equipotential regions and counted points or lines of equilibrium according to their degrees. He asserted, without proof, that for charged points or spherical conductors the number of additional equilibria arising through cyclosis cannot exceed $(n-1)(n-2)$, where n is the number of bodies [9, §113]. This estimate already appears in the 1873 edition. The third edition, edited by J. J. Thomson and published in 1892, adds Thomson's bracketed footnote stating that he had been unable to find a proof of the estimate [10, §113]. Gabrielov, Novikov, and Shapiro explained how Maxwell's weighted counts and this auxiliary estimate lead to the bound $(n-1)^2$, and formulated the modern conjecture for configurations whose equilibria are all nondegenerate [5, Conjecture 1.3 and §4].

For the generalized potentials considered in their paper, Gabrielov, Novikov, and Shapiro proved the upper bound 12 for nondegenerate three-charge configurations; in particular, this applies to the Coulomb potential in $\mathbb{R}^3$ [5, Theorem 1.5(b)]. Tsai proved the four-equilibrium bound for the planar problem when the three sites form an equilateral or isosceles right triangle [13]. He later treated arbitrary three-site configurations with equal charge magnitudes: when the critical set is finite, the number of isolated equilibria is at most four, and the exceptional configurations with infinitely many equilibria are classified [14]. Related planar approaches appear in [8, 11].

Zolotov obtained the general bound $5 \cdot 9^{n+3}$ for the number of isolated Coulomb critical points [16]. Edelsbrunner, Fillmore, and Oliveira subsequently obtained the bound $2^n(3n-2)^3$ for arbitrary nonzero strengths when one charge site is in generic position [4, Result 1]. Arathoon, Ball, and Kvalheim constructed five positive charges whose potential has at least 24 nondegenerate critical points. They further showed that an arbitrarily small perturbation of the strengths gives a finite critical set of cardinality at least 24, all of whose points are nondegenerate, thereby disproving the general Maxwell conjecture [2, Theorem 1]. As of [2, §1], the three-charge problem was not known beyond Tsai's equal-magnitude cases. Accordingly, to the best of our knowledge, the result below

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is the first proof of the sharp four-point bound for arbitrary distinct three-site configurations and arbitrary nonzero real strengths under the finite-critical-set hypothesis.

Theorem 1 (Three-charge Maxwell bound)**.** *Let $A = \{a_1, a_2, a_3\} \subset \mathbb{R}^3$ consist of three distinct points, let $q \in (\mathbb{R} \setminus \{0\})^3$, and define*

$$V_q : \mathbb{R}^3 \setminus A \longrightarrow \mathbb{R}, \qquad V_q(x) = \sum_{i=1}^{3} \frac{q_i}{|x - a_i|}.$$

If

$$\operatorname{Crit}(V_q) := \{x \in \mathbb{R}^3 \setminus A : \nabla V_q(x) = 0\}$$

is finite, then $\# \operatorname{Crit}(V_q) \leq 4$, the points being counted without multiplicity; see Remark 1.1 for sharpness. Moreover, $\# \operatorname{Crit}(V_q) \leq 2$ if the sites are collinear, and $\# \operatorname{Crit}(V_q) \leq 3$ if the sites are noncollinear and $\sum_i q_i = 0$.

Remark 1.1 (Sharpness)*.* The constant 4 is sharp: three equal positive charges at the vertices of an equilateral triangle have four nondegenerate equilibria [14, 2]. The collinear constant 2 is sharp: three equal positive charges on a line have exactly one equilibrium in each of the two gaps. The constant 3 in the noncollinear zero-sum case is, by contrast, almost certainly *not* attained: Remark 9.3 reduces the question to a single explicit inequality, proves part of it, and records the computational evidence that a noncollinear zero-sum configuration always has exactly one equilibrium.

The finiteness hypothesis is necessary for a literal cardinality statement with signed charges. For example, take sites

$$a_1 = (-1, 0, 0), \qquad a_2 = (0, 0, 0), \qquad a_3 = (1, 0, 0)$$

and strengths $(-2\sqrt{2}, 2, -2\sqrt{2})$. At every point $x = (0, y, z)$ with $y^2 + z^2 = 1$, the three coefficients $q_i/|x - a_i|^3$ are $(-1, 2, -1)$; hence $\sum_i q_i(x - a_i)/|x - a_i|^3 = 0$. Thus the entire unit circle is critical, so the finiteness hypothesis cannot be omitted for arbitrary signed strengths. The theorem makes no assertion about configurations with positive-dimensional critical sets, and the hypothesis is weaker than nondegeneracy because isolated degenerate critical points are allowed. For the modern nondegenerate formulation, finiteness is automatic. Indeed, introduce variables $r_i > 0$ with $r_i^2 = |x - a_i|^2$. After multiplication by $\prod_i r_i^3$, the equations $\nabla V_q(x) = 0$ are polynomial in (x, r_1, r_2, r_3). The Tarski–Seidenberg theorem therefore makes the critical set semialgebraic. Nondegeneracy makes each of its points isolated, while a semialgebraic set has only finitely many connected components; hence it cannot contain infinitely many isolated points.

1.1 Outline

The noncollinear proof occupies Sections 2–9 and has four stages. First, a projective charge map Θ identifies equilibria with fibres over charge rays, and a complex shape function Ξ identifies the singular locus of Θ (Section 2). Second, the region $|\Xi| < 1/3$ is shown to consist of four discs, one in each allowed barycentric sign chamber (Section 3). Third, an exact Lami identity proves that the dual normal of each caustic turns monotonically (Sections 4–6); a fourth-order calculation then gives exactly three ordinary cusps (Section 7), and an anti-periodic support-envelope lemma makes each caustic a Jordan curve (Section 8). Finally, Brouwer degree and the planar gradient index formula give the bound (Section 9). Collinear sites are handled separately in Section 9.1 by zonal harmonic expansions.

The longer polynomial identities are computer-assisted but exact. Every such identity is reconstructed from displayed definitions; the polynomials $\mathcal{V}$ and $\mathcal{S}$ that carry them are constructed explicitly in Section 3 and recorded in Appendices A–B, and the complete assertion-based certificate suite is described in Section 10.

1.2 Notation

Throughout, $A = \{a_1, a_2, a_3\}$ and $q = (q_1, q_2, q_3)$ with every $q_i \neq 0$. When the sites are noncollinear we write P for their affine plane, identified with $\mathbb{C}$, and

$$P_A = P \setminus A, \qquad u_i = x - a_i = r_i e^{i\alpha_i}, \qquad e_i = \frac{u_i}{\bar{u}_i} = e^{2i\alpha_i}.$$

We fix once and for all the labelling and the orientation of P so that the signed area

$$S = \tfrac{1}{2}\det(a_2 - a_1,\, a_3 - a_1)$$

is positive; this is possible after a cyclic relabelling, and every relabelling used below is cyclic, hence orientation preserving. All gradients, Hessians, indices and winding numbers are two-dimensional unless stated otherwise.

The following symbols are used with a fixed meaning throughout.

λ_i	normalized barycentrics of x	$\Xi = \sum_i \lambda_i e_i$	shape function
$\vartheta(x) = (\lambda_i r_i^3)_i$	canonical charge	$\Theta = [\vartheta]$	projective charge map
$\mathcal{R} = \{\lvert\Xi\rvert < 1/3\}$	index-+1 region	Ω	one component of $\mathcal{R}$
$T = \operatorname{conv} A$, $\widehat{T}$	triangle, its blow-up	φ_i	interior angle of T at a_i
L_i, $\ell_i = \sqrt{L_i}$	squared side lengths, sides	y_i	Ravi variables
$\mathcal{V}$, $\mathcal{T}$	interior Jacobian, AM–GM tail	(s, t)	edge-chamber chart
$\mathcal{Q}_i$, $\mathcal{Q}$	$(\sum \hat{\lambda})^2 r_i^2$, their product	Φ, $\mathcal{L}$	$\mathcal{H}$-numerator, fold equation
$\mathcal{H} = 1 - \lvert\Xi\rvert^2$		$\mathcal{E}$, $\mathcal{S}$	edge-fold eliminants
ψ, $\beta_i = \alpha_i - \psi/2$	fold phase, reduced angles	$t_i = \tan \beta_i$	tangent chart on the fold
$n, \widehat{n}, \nu$	Lami covectors, charge vector	$\mathfrak{a}, \mathfrak{b}$	Lami coefficients
σ, D_σ, E	chamber signs, chart data	$\mathbf{N}$, c_{aff}	affine dual line
h, $\rho = h + h''$, Γ	support function, envelope	θ	normal angle
θ_i, c_i, s_i	directions $x_0 \to a_i$, cosine, sine	$\mathfrak{d}_i$, $\mathfrak{D}$	$\sin(\theta_k - \theta_j)$, their product
$\mathfrak{p}_i$	$2c_j c_k + s_j s_k$	$\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$	elementary symmetric functions
$\mathfrak{N}$, $\mathfrak{s}$, m, d	equilibrium counts	w_∞	winding at infinity

2 Planarity and the charge map

Throughout this section the sites are noncollinear.

Lemma 2.1 (Planarity)**.** $\operatorname{Crit}(V_q) \subset P$*, and* $\operatorname{Crit}(V_q)$ *is exactly the zero set of* $\nabla_P V_q$ *on* P_A*, where* ∇_P *denotes the gradient of the restriction* $V_q|_P$*.*

Proof. Write $x = (x_P, z)$ after choosing $P = \{z = 0\}$, and set $m_i = q_i/|x - a_i|^3$, so that $\nabla V_q(x) = -\sum_i m_i(x - a_i)$. If $\nabla V_q(x) = 0$ and $z \neq 0$, the normal component gives $\sum_i m_i = 0$ because every a_i has vanishing normal coordinate; the tangential component then gives $\sum_i m_i a_i = (\sum_i m_i) x_P = 0$. Thus (m_1, m_2, m_3) is an affine dependence among a_1, a_2, a_3. The sites are affinely independent in P, so every m_i vanishes, contrary to $q_i \neq 0$. Hence every equilibrium lies in P. Conversely, V_q is invariant under the reflection fixing P, so its normal derivative vanishes identically on P; therefore the zeros of $\nabla_P V_q$ are exactly the three-dimensional equilibria. ∎

For $x \in P_A$ let $\lambda_i = \lambda_i(x)$ be the normalized barycentric coordinates, characterized by

$$\sum_i \lambda_i = 1, \qquad \sum_i \lambda_i u_i = 0,$$

equivalently $x = \sum_i \lambda_i a_i$; they exist and are unique because the sites are affinely independent. Since $\sum_i \lambda_i = 1$, the vector λ never vanishes, and neither does

$$\vartheta(x) = (\lambda_1 r_1^3, \lambda_2 r_2^3, \lambda_3 r_3^3).$$

Lemma 2.2 (Charge map)**.** *The map $\Theta : P_A \to \mathbb{RP}^2$, $\Theta(x) = [\vartheta(x)]$, is well defined, and*

$$x \in \mathrm{Crit}(V_q) \iff [q] = \Theta(x) = [\lambda_1 r_1^3 : \lambda_2 r_2^3 : \lambda_3 r_3^3]. \tag{2.1}$$

Proof. Criticality means $\sum_i (q_i/r_i^3)u_i = 0$. The linear map $\mathbb{R}^3 \to P$, $c \mapsto \sum_i c_i u_i$, has rank two: its rank could drop only if all three u_i were parallel, which would force x and the three sites to be collinear. Its kernel is therefore one dimensional, and it contains the nonzero vector λ; hence the kernel is $\mathbb{R}\lambda$. Thus $\sum_i (q_i/r_i^3)u_i = 0$ if and only if $q_i = \varsigma \lambda_i r_i^3$ for some $\varsigma \neq 0$. □

Define

$$\Xi(x) = \sum_i \lambda_i e_i = \sum_i \lambda_i e^{2i\alpha_i}, \qquad \mathcal{R} = \{x \in P_A : |\Xi(x)| < 1/3\}.$$

Lemma 2.3 (Hessian and singular locus)**.** *For the canonical representative $q = \vartheta(x)$ the planar Hessian of V_q at x is*

$$H_0(x) = \tfrac{1}{2}\big(I + 3\mathsf{M}(\Xi(x))\big), \qquad \mathsf{M}(z) = \begin{pmatrix} \Re z & \Im z \\ \Im z & -\Re z \end{pmatrix}. \tag{2.2}$$

If the physical charge vector is $q = \varsigma\,\vartheta(x)$ with $\varsigma \neq 0$, then the full Hessian of V_q at x in $P \oplus P^\perp$ is $\varsigma H_0(x) \oplus (-\varsigma)$. Moreover

$$H_0(x) + \overline{K}_x\, d\Theta_x = 0, \qquad d\Theta_x = -\overline{K}_x^{-1} H_0(x), \tag{2.3}$$

where $\overline{K}_x$ is induced by the charge-variation map $K_x(\dot q) = -\sum_i (\dot q_i/r_i^3)u_i$, and consequently

$$\operatorname{rank} d\Theta_x < 2 \iff |\Xi(x)| = 1/3, \tag{2.4}$$

with $d\Theta_x$ of rank one on this level.

Proof. For a single source, $\nabla_P^2(|u|^{-1}) = (3uu^\mathsf{T} - |u|^2 I)/|u|^5$ on P. Hence, with $q_i = \lambda_i r_i^3$,

$$H_0 = \sum_i \lambda_i\Big(3\frac{u_i u_i^\mathsf{T}}{r_i^2} - I\Big) = 3\sum_i \lambda_i \widehat{u}_i \widehat{u}_i^\mathsf{T} - I, \qquad \widehat{u}_i = u_i/r_i.$$

For a unit vector at angle α one has $\widehat{u}\widehat{u}^\mathsf{T} = \frac{1}{2}(I + \mathsf{M}(e^{2i\alpha}))$, so $H_0 = \frac{3}{2}I + \frac{3}{2}\mathsf{M}(\Xi) - I$, which is (2.2). The matrix $\mathsf{M}(z)$ is symmetric and traceless with eigenvalues $\pm|z|$, so H_0 has eigenvalues $(1 \pm 3|\Xi|)/2$ and unit trace. Since V_q is harmonic and the mixed derivatives vanish on P by reflection symmetry, the normal second derivative equals $-\varsigma$, which gives the stated block form.

The planar eigenvalues of the physical Hessian are $\varsigma(1 \pm 3|\Xi|)/2$, so the planar determinant has the sign of $1 - 9|\Xi|^2$ independently of ς. Thus an equilibrium in $\mathcal{R}$ is nondegenerate of planar index $+1$, one outside $\overline{\mathcal{R}}$ is a regular saddle of planar index -1, and one on $\partial\mathcal{R}$ is degenerate.

The map K_x is exactly $\dot q \mapsto \nabla_P V_{\dot q}(x)$; by the proof of Lemma 2.2 its kernel is $\mathbb{R}\vartheta(x)$, so it induces an isomorphism $\overline{K}_x : \mathbb{R}^3/\mathbb{R}\vartheta(x) \to P$, and we identify $T_{\Theta(x)}\mathbb{RP}^2$ with $\mathbb{R}^3/\mathbb{R}\vartheta(x)$ by means of the representative $\vartheta(x)$. The incidence equation $\nabla_P V_{\vartheta(x)}(x) = -\sum_i \lambda_i u_i = 0$ holds identically in x; differentiating it gives (2.3). Since $\overline{K}_x$ is an isomorphism, $\operatorname{rank} d\Theta_x = \operatorname{rank} H_0(x)$, and $\det H_0$ has the sign of $1 - 9|\Xi|^2$, whence (2.4); on that level H_0 has eigenvalues 0 and 1, so the rank is one. □

A signed Jacobian of Θ will only be used after choosing an oriented affine chart in a fixed charge sign chamber; no global orientation of $\mathbb{RP}^2$ is being assumed.

Lemma 2.4 (Localization)**.** *Let O and R_c be the circumcenter and circumradius of T. Then*

$$1 - |\Xi|^2 = \frac{16S^2\lambda_1\lambda_2\lambda_3(R_c^2 - |x-O|^2)}{r_1^2r_2^2r_3^2}. \tag{2.5}$$

Consequently $\overline{\mathcal{R}}$ meets exactly four barycentric sign chambers, namely $(+++)$, which is $\operatorname{int} T$, and the three chambers with a single negative coordinate, in each of which $\overline{\mathcal{R}}$ lies outside the circumcircle. Moreover, if $\Theta(x) = \Theta(x')$ with $x, x' \in \overline{\mathcal{R}}$, then x and x' lie in the same closed sign chamber.

Proof. Since $|e_i| = 1$ and $\sum_i \lambda_i = 1$, expanding both sides gives the elementary identity $1 - \left|\sum_i \lambda_i e_i\right|^2 = \sum_{i<j} \lambda_i\lambda_j|e_i - e_j|^2$. Next, $e_i - e_j = (u_i\bar u_j - u_j\bar u_i)/(\bar u_i \bar u_j)$ and $\Im(u_i\bar u_j) = -\det(u_i, u_j) = -2S\lambda_k$, so $|e_i - e_j|^2 = 16S^2\lambda_k^2/(r_i^2r_j^2)$. Finally $\sum_i \lambda_i r_i^2 = R_c^2 - |x-O|^2$, because expanding $|x-a_i|^2$ about O and using $\sum_i \lambda_i a_i = x$ leaves $|x-O|^2 - 2|x-O|^2 + R_c^2$. Combining the three displays gives (2.5).

No point of $\overline{\mathcal{R}}$ has a vanishing barycentric coordinate: on $L_{ij} \setminus \{a_i, a_j\}$, where $L_{ij} = \operatorname{aff}\{a_i, a_j\}$, one has $\lambda_k = 0$ and $e_i = e_j$, hence $|\Xi| = 1$. No point of $\overline{\mathcal{R}}$ lies in a chamber with two negative coordinates: if $\lambda_i > 0 > \lambda_j, \lambda_k$ then the reverse triangle inequality gives

$$|\Xi| \ge \lambda_i - |\lambda_j| - |\lambda_k| = \lambda_i + \lambda_j + \lambda_k = 1.$$

The pattern $(---)$ is impossible because $\sum_i \lambda_i = 1$. This leaves the four asserted chambers, and (2.5) together with $1 - |\Xi|^2 > 0$ forces $|x - O| > R_c$ in each one-negative chamber.

For the last statement, $\Theta(x) = \Theta(x')$ means $\lambda_i(x)r_i(x)^3 = \varsigma\,\lambda_i(x')r_i(x')^3$ for all i and some $\varsigma \neq 0$, so the two sign patterns are equal or opposite. None of the four admissible patterns is the negative of another, so they are equal. □

Section 3 shows that the portion of $\overline{\mathcal{R}}$ in each admissible chamber is a closed disc.

3 The four elliptic discs

Throughout this section we use the normalization

$$a_1 = 0, \qquad a_2 = 1, \qquad a_3 = \xi_0 + i\eta_0, \qquad \eta_0 > 0, \tag{3.1}$$

so that $2S = \eta_0 > 0$ and the complex orientation of the x-plane is the chosen one. We write $L_1 = |a_2 - a_3|^2$, $L_2 = |a_3 - a_1|^2$, $L_3 = |a_1 - a_2|^2$ and $\ell_i = \sqrt{L_i}$, and we let φ_i be the interior angle of T at a_i.

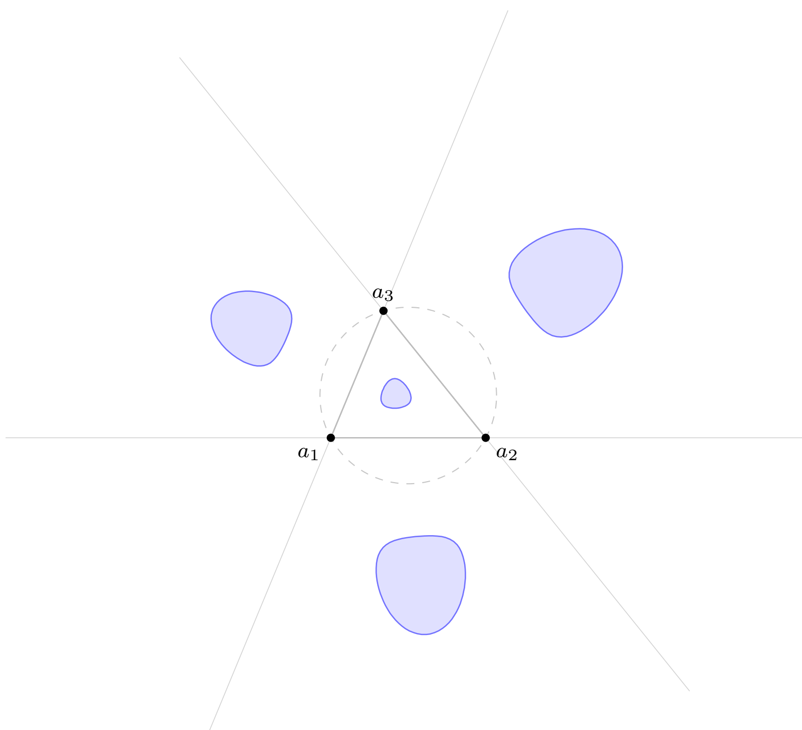

Figure 1: The region $\mathcal{R} = \{|\Xi| < 1/3\}$ for the triangle $a_1 = 0$, $a_2 = 1$, $a_3 = 0.34 + 0.82\,i$. It consists of four open discs, one in each admissible barycentric sign chamber: the chamber $(+++)$ inside the triangle, and the three chambers with a single negative coordinate, which lie across the corresponding side and, by Lemma 2.4, outside the circumcircle (dashed). The site lines are shown in grey.

3.1 Boundary behaviour of Ξ

We first record, once and for all, the compactification used in both chambers. Let $\mathcal{S}_0$ be a compact set of normalized shapes (ξ_0, η_0) in the open upper half plane.

Lemma 3.1 (Boundary behaviour)**.** *Uniformly for $(\xi_0, \eta_0) \in \mathcal{S}_0$:*

(i) *On an open side $\lambda_i = 0$ one has $\Xi = e_j = e_k$; this value is constant along the side and has modulus one.*

(ii) *If $x - a_i = re^{i\theta}$ with $r \to 0$, then $\Xi = e^{2i\theta} + O(r)$, uniformly in θ.*

(iii) *As $|x| \to \infty$, $\Xi(x) = x/\bar{x} + O(|x|^{-1})$, uniformly in $\arg x$.*

Consequently Ξ extends continuously to the compact surface obtained from the closed chamber by real blowing up the finite vertices and adjoining the circle at infinity, jointly continuously in $(\xi_0, \eta_0) \in \mathcal{S}_0$, with $|\Xi| = 1$ on every boundary face. In particular, for every $\epsilon > 0$ there is a compact subset of the open chamber, varying continuously with the shape, outside of which $|\Xi| > 1 - \epsilon$.

Proof. (i) If $\lambda_i = 0$ then $\lambda_j u_j + \lambda_k u_k = 0$ with $\lambda_j + \lambda_k = 1$, so u_j and u_k are parallel. Since $u \mapsto u/\bar{u}$ is invariant under multiplication of u by a nonzero real, $e_j = e_k$ and $\Xi = \lambda_j e_j + \lambda_k e_k = e_j$. Along the side $a_j a_k$ the direction of u_j is constant, so Ξ is constant of modulus one.

(ii) Cramer's rule gives $\lambda_j, \lambda_k = O(r)$ with constants bounded in terms of $1/\eta_0$ and $\max_{p,q} |a_p - a_q|$, hence uniformly on $\mathcal{S}_0$, while $\lambda_i \to 1$ and $|e_p| = 1$. Therefore $\Xi = e_i + O(r) = e^{2i\theta} + O(r)$.

(iii) Expanding,

$$e_i = \frac{x}{\bar{x}} + \frac{x\bar{a}_i - \bar{x}a_i}{\bar{x}^2} + O(|x|^{-2}).$$

Because $\sum_i \lambda_i = 1$ and $\sum_i \lambda_i a_i = x$, the first-order terms cancel exactly after summation: $\sum_i \lambda_i (x\bar{a}_i - \bar{x}a_i)/\bar{x}^2 = (x\bar{x} - \bar{x}x)/\bar{x}^2 = 0$. Since $\lambda_i = O(|x|)$ with constants uniform on $\mathcal{S}_0$, the remaining terms contribute $O(|x|^{-1})$.

For the compactification statement, consider the edge chamber; the interior chamber is the same argument with no face at infinity. In the chart of Section 3.3 the closed square $[0,1]^2$ compactifies the chamber: the sides $s = 0$ and $s = 1$ are the two open sides of the chamber, the side $t = 0$ is the segment $a_2 a_3$, the corners $(0,0)$ and $(1,0)$ correspond to the sites a_3 and a_2, and $t = 1$ is the face at infinity. Near a site corner the chart is an orientation-preserving diffeomorphism onto a sector at that site, so the real blow-up of the corner matches the real blow-up of P at the site, and Ξ extends by (ii). On the face $t = 1$ the limit is $x/\bar{x}$ by (iii), with $\arg x$ the direction of the chamber ray determined by s; this depends continuously on s. At the two remaining corners $(0,1)$ and $(1,1)$ the two one-sided limits agree, because the chamber ray in direction $s \to 0$ is the ray of L_{13} beyond a_3 and that in direction $s \to 1$ is the ray of L_{12} beyond a_2, whose squared directions are precisely the constant values of Ξ on the sides $s = 0$ and $s = 1$ given by (i). All the estimates above are uniform on $\mathcal{S}_0$, so Ξ is continuous on the compact product of the compactified chamber with $\mathcal{S}_0$, with modulus one on every boundary face. The last statement follows by compactness. □

3.2 Interior chamber

We first construct the polynomial that controls the Jacobian. Treat z and $w = \bar{z}$ as independent variables and put, for cyclic (i, j, k),

$$u_i = z - a_i, \qquad v_i = w - \bar{a}_i, \qquad D_i = v_j u_k - u_j v_k, \qquad c_i = a_k - a_j.$$

Then $\sum_i D_i = 2i\eta_0$, the barycentrics are $\lambda_i = D_i/(2i\eta_0)$, and $\Xi = \sum_i \lambda_i u_i/v_i$. Exact Wirtinger differentiation gives

$$\Xi_z = \frac{\Pi}{2i\eta_0(v_1v_2v_3)^2}, \qquad \Xi_{\bar z} = \frac{\Pi^\sharp}{2i\eta_0(v_1v_2v_3)^2},$$

$$\Pi = \sum_{\mathrm{cyc}} (\bar c_i u_i + D_i) v_i v_j^2 v_k^2, \qquad \Pi^\sharp = -\sum_{\mathrm{cyc}} u_i(c_i v_i + D_i) v_j^2 v_k^2.$$

Since $\operatorname{Jac}\Xi = |\Xi_z|^2 - |\Xi_{\bar z}|^2$ and $|v_i| = r_i$, the difference $|\Pi^\sharp|^2 - |\Pi|^2$, expressed in the barycentric variables, is exactly divisible by $\lambda_1\lambda_2\lambda_3\eta_0^4$, and we *define*

$$|\Pi^\sharp|^2 - |\Pi|^2 = \lambda_1\lambda_2\lambda_3\,\eta_0^4\,\mathcal{V}. \tag{3.2}$$

Then

$$\operatorname{Jac}\Xi = \frac{|\Pi|^2 - |\Pi^\sharp|^2}{4\eta_0^2\,r_1^4r_2^4r_3^4} = -\,\frac{S^2\lambda_1\lambda_2\lambda_3}{r_1^4r_2^4r_3^4}\,\mathcal{V}, \tag{3.3}$$

which is manifestly invariant under similarities. As constructed, $\mathcal{V}$ is a polynomial in $(\lambda_1, \lambda_2, \lambda_3)$, homogeneous of degree 11, whose coefficients are polynomials in ξ_0, η_0 involving only even powers of η_0. Substituting $\eta_0^2 = L_2 - \xi_0^2$ and $\xi_0 = (1 + L_2 - L_1)/2$, and then homogenizing with L_3 (which equals 1 in the normalization (3.1)), expresses $\mathcal{V}$ as a polynomial

$$\mathcal{V} = \mathcal{V}(\lambda_1, \lambda_2, \lambda_3; L_1, L_2, L_3),$$

bihomogeneous of degree 11 in the λ_i and degree 3 in the L_i, and invariant under every simultaneous permutation of the paired variables (λ_i, L_i). All of this is verified exactly in `certificates/interior_jacobian.py`; see Appendix A.

Proposition 3.2 (Interior disc)**.** *$\operatorname{Jac}\Xi < 0$ throughout $\operatorname{int} T$, the map $\Xi : \operatorname{int} T \to \mathbb{D} = \{|z| < 1\}$ is an orientation-reversing diffeomorphism, and $\Xi^{-1}\{|z| \le 1/3\}$ is a closed disc contained in $\operatorname{int} T$ whose boundary $\Xi^{-1}\{|z| = 1/3\}$ is one smooth circle.*

Proof. Put $y_i = (\ell_j + \ell_k - \ell_i)/2$, which is positive by the strict triangle inequality, and apply the Ravi substitution $L_i = (y_j + y_k)^2$. With $X = (\lambda_1, \lambda_2, \lambda_3, y_1, y_2, y_3)$ the resulting polynomial $\mathcal{V}_{\mathrm{Ravi}}$ is bihomogeneous of degree $(11, 6)$ with integer coefficients, and it admits the exact decomposition

$$\mathcal{V}_{\mathrm{Ravi}} = \sum_{r=1}^{30} \frac{\mathsf{n}_r}{2}\left(X^{\alpha_r} + X^{\gamma_r} - 2X^{\beta_r}\right) + \mathcal{T}(X), \qquad \alpha_r + \gamma_r = 2\beta_r, \tag{3.4}$$

where $\mathsf{n}_r \in \mathbb{Z}_{>0}$, $\alpha_r, \beta_r, \gamma_r \in \mathbb{Z}_{\ge 0}^6$, and $X^\alpha = \prod_{j=1}^6 X_j^{\alpha_j}$. The thirty quadruples $(\mathsf{n}_r, \alpha_r, \gamma_r, \beta_r)$ are listed in Table 1; the β_r are precisely the thirty monomials of $\mathcal{V}_{\mathrm{Ravi}}$ with a negative coefficient. Each binomial in (3.4) is nonnegative for $X > 0$ by the AM–GM inequality, because $\alpha_r + \gamma_r = 2\beta_r$. The remainder $\mathcal{T}$ has 1548 terms, all of its coefficients are positive integers, and the smallest of them is 16; in particular $\mathcal{T}(X) > 0$ for $X > 0$. Hence $\mathcal{V}_{\mathrm{Ravi}} > 0$ for positive X, so $\mathcal{V} > 0$ for positive barycentrics, and (3.3) gives $\operatorname{Jac}\Xi < 0$ on $\operatorname{int} T$.

Every point of $\operatorname{int} T$ lies strictly inside the circumdisc, so (2.5) gives $|\Xi| < 1$ there. By Lemma 3.1, Ξ extends continuously to the compact surface $\widehat{T}$ obtained by blowing up the three vertices, with modulus one on $\partial\widehat{T}$. For every compact $K \Subset \mathbb{D}$ the set $\Xi^{-1}(K)$ is closed in $\widehat{T}$ and disjoint from $\partial\widehat{T}$, hence compact in $\operatorname{int} T$; therefore

$$\Xi : \operatorname{int} T \longrightarrow \mathbb{D}$$

is proper, and its Brouwer degree is defined and equals the degree of the boundary map $\partial\widehat{T} \to S^1$. By Lemma 3.1(i) the side pieces of $\partial\widehat{T}$ map to constants. Traversing $\partial\widehat{T}$ positively, one enters the blow-up arc at a_i along one side and leaves along the other, so $\arg u_i$ decreases by the interior angle φ_i and $\arg \Xi$ decreases by $2\varphi_i$. Thus the boundary degree is $-2(\varphi_1 + \varphi_2 + \varphi_3)/(2\pi) = -1$. Since every local Jacobian is negative, for every $y \in \mathbb{D}$

$$-1 = \deg(\Xi, \operatorname{int} T, y) = -\#\Xi^{-1}(y),$$

so Ξ is a bijection onto $\mathbb{D}$; being also a local diffeomorphism, it is a diffeomorphism. The last assertion follows at once. □

3.3 Edge chambers

By the cyclic relabelling fixed in Section 1.2 it is enough to consider $\lambda_1 < 0 < \lambda_2, \lambda_3$. Use the homogeneous barycentric coordinates

$$(\hat\lambda_1 : \hat\lambda_2 : \hat\lambda_3) = (-t : s : 1-s), \qquad 0 < s, t < 1;$$

the normalized coordinates are $(-t, s, 1-s)/(1-t)$. Conversely, for normalized barycentrics with $\lambda_1 < 0 < \lambda_2, \lambda_3$, set

$$s = \frac{\lambda_2}{\lambda_2 + \lambda_3}, \qquad t = \frac{-\lambda_1}{\lambda_2 + \lambda_3}.$$

Then $0 < s, t < 1$, $\lambda = (-t, s, 1-s)/(1-t)$, and

$$\det \frac{\partial(\lambda_2, \lambda_3)}{\partial(s, t)} = (1-t)^{-3} > 0.$$

Since $S > 0$, the affine chart $(\lambda_2, \lambda_3) \mapsto x$ is orientation preserving; hence so is (s, t), which is therefore a global orientation-preserving chart of the strict edge chamber. Put

$$K_{\mathrm{e}} = L_1 s(1-s), \qquad D_{\mathrm{e}} = L_2(1-s) + L_3 s, \qquad P_{\mathrm{e}} = D_{\mathrm{e}} t - K_{\mathrm{e}},$$

and, with $(\hat\lambda_1, \hat\lambda_2, \hat\lambda_3) = (-t, s, 1-s)$,

$$\begin{aligned}
\mathcal{Q}_1 &= L_3\hat\lambda_2^2 + (L_2 + L_3 - L_1)\hat\lambda_2\hat\lambda_3 + L_2\hat\lambda_3^2,\\
\mathcal{Q}_2 &= L_1\hat\lambda_3^2 + (L_3 + L_1 - L_2)\hat\lambda_3\hat\lambda_1 + L_3\hat\lambda_1^2,\\
\mathcal{Q}_3 &= L_2\hat\lambda_1^2 + (L_1 + L_2 - L_3)\hat\lambda_1\hat\lambda_2 + L_1\hat\lambda_2^2.
\end{aligned}$$

Expanding $x - a_i$ in the two edge vectors at a_i gives $\mathcal{Q}_i = (\hat\lambda_1 + \hat\lambda_2 + \hat\lambda_3)^2 r_i^2 > 0$. Writing $\mathcal{H} = 1 - |\Xi|^2$ and using Heron's formula $2L_1L_2 + 2L_1L_3 + 2L_2L_3 - L_1^2 - L_2^2 - L_3^2 = 16S^2$, direct substitution in (2.5) gives

$$\mathcal{H} = \frac{\Phi}{\mathcal{Q}}, \qquad \Phi = 16S^2\, ts(1-s)P_{\mathrm{e}}(1-t), \qquad \mathcal{Q} = \mathcal{Q}_1\mathcal{Q}_2\mathcal{Q}_3, \tag{3.5}$$

the key intermediate identity being $\hat\lambda_1\mathcal{Q}_1 + \hat\lambda_2\mathcal{Q}_2 + \hat\lambda_3\mathcal{Q}_3 = -(1-t)P_{\mathrm{e}}$. Thus the fold equation is

$$\mathcal{L} := 9\Phi - 8\mathcal{Q} = 0. \tag{3.6}$$

On the fold $\{\mathcal{L} = 0\}$ one has $\mathcal{H} = 8/9 > 0$, so (3.5) gives $\Phi > 0$ and therefore $P_{\mathrm{e}} > 0$.

Proposition 3.3 (Edge fold is regular)**.** *In every edge chamber,*

$$|\Xi| = 1/3 \quad \Longrightarrow \quad d|\Xi|^2 \neq 0. \tag{3.7}$$

Proof. On the fold (3.6) we have $\mathcal{H} = \Phi/\mathcal{Q}$ with $9\Phi = 8\mathcal{Q}$, so for $j \in \{s,t\}$

$$\mathcal{H}_j = \frac{\Phi_j \mathcal{Q} - \Phi \mathcal{Q}_j}{\mathcal{Q}^2} = \frac{9\Phi_j - 8\mathcal{Q}_j}{9\mathcal{Q}} = \frac{\Phi\,\mathcal{L}_j}{8\mathcal{Q}^2} \qquad \text{on } \{\mathcal{L} = 0\}.$$

The numerator $\Phi_t\mathcal{Q} - \Phi\mathcal{Q}_t$ of $\mathcal{H}_t$ factors exactly, as an identity of polynomials, as

$$\Phi_t\mathcal{Q} - \Phi\mathcal{Q}_t = -16S^2 s(1-s)\,\mathcal{Q}_1\mathcal{G}_1\mathcal{G}_2, \tag{3.8}$$

$$\mathcal{G}_1 = K_{\mathrm{e}}(1-2t) + D_{\mathrm{e}}t^2, \qquad \mathcal{G}_2 = P_{\mathrm{e}}^2 - \ell_2^2\ell_3^2 t^2(1-t)^2.$$

If $\mathcal{G}_1 = 0$ then $D_{\mathrm{e}}t^2 = K_{\mathrm{e}}(2t-1)$ forces $t > 1/2$, whence $P_{\mathrm{e}} = K_{\mathrm{e}}(t-1)/t < 0$, a contradiction. Moreover

$$\mathcal{G}_2 = \mathcal{G}_0\big(P_{\mathrm{e}} + \ell_2\ell_3 t(1-t)\big), \qquad \mathcal{G}_0 := P_{\mathrm{e}} - \ell_2\ell_3 t(1-t),$$

and the second factor is positive. Hence, at a fold point with $d\mathcal{H} = 0$, we get $\mathcal{G}_0 = 0$ from $\mathcal{H}_t = 0$ and $\mathcal{L}_s = 0$ from $\mathcal{H}_s = 0$, so that

$$\mathcal{L} = \mathcal{L}_s = \mathcal{G}_0 = 0.$$

We eliminate t. In the strict chamber the leading coefficients in t do not vanish:

$$\mathrm{lc}_t\,\mathcal{L} = -8\ell_2^2\ell_3^2\,\mathcal{Q}_1, \qquad \mathrm{lc}_t\,\mathcal{G}_0 = \ell_2\ell_3,$$

so the resultant computed over $\mathbb{Q}(\ell_1,\ell_2,\ell_3,s)[t]$ specializes faithfully. Exact elimination gives

$$\mathrm{Res}_t(\mathcal{L},\mathcal{G}_0) = s^2(1-s)^2\ell_1^4\ell_2^2\ell_3^2(\ell_1-\ell_2+\ell_3)^2(\ell_1+\ell_2-\ell_3)^2\mathcal{Q}_1^2\,\mathcal{E}^2, \tag{3.9}$$

where

$$\begin{aligned}\mathcal{E} = \ell_1^2s^2 - \ell_1^2 s - 9\ell_2^2s^2 + 17\ell_2^2 s - 8\ell_2^2 \\ - 18\ell_2\ell_3s^2 + 18\ell_2\ell_3 s - 9\ell_3^2s^2 + \ell_3^2 s.\end{aligned}$$

Next, $\mathrm{Res}_t(\mathcal{L}_s,\mathcal{G}_0)$ is exactly divisible by $s(s-1)\ell_1^4\ell_2\ell_3(\ell_1-\ell_2+\ell_3)^2(\ell_1+\ell_2-\ell_3)^2\mathcal{Q}_1$, and we define $\mathcal{S} \in \mathbb{Z}[\ell_1,\ell_2,\ell_3,s]$ by the exact quotient

$$\mathrm{Res}_t(\mathcal{L}_s,\mathcal{G}_0) = s(s-1)\ell_1^4\ell_2\ell_3(\ell_1-\ell_2+\ell_3)^2(\ell_1+\ell_2-\ell_3)^2\mathcal{Q}_1\,\mathcal{S}. \tag{3.10}$$

The quotient has zero remainder, so this specifies $\mathcal{S}$ uniquely; it has degree 6 in s and total degree 14, and is displayed in Appendix B. For (3.10) we use only the elementary implication that two polynomials with a common root have singular Sylvester matrix, whatever formal degrees are used; no hypothesis on $\mathrm{lc}_t\,\mathcal{L}_s$ is needed.

All displayed factors other than $\mathcal{E}$ and $\mathcal{S}$ are nonzero in the strict chamber, so $\mathcal{L} = \mathcal{L}_s = \mathcal{G}_0 = 0$ forces $\mathcal{E} = \mathcal{S} = 0$ at the same value of s. Eliminating s gives

$$\begin{aligned}\mathrm{Res}_s(\mathcal{E},\mathcal{S}) = 5184\,\ell_2^6\ell_3^6(\ell_1-3\ell_2-3\ell_3)^2(\ell_1-\ell_2-\ell_3)^4(\ell_1+\ell_2+\ell_3)^4 \\ \cdot(\ell_1+3\ell_2+3\ell_3)^2(\ell_1^2-\ell_2^2-34\ell_2\ell_3-\ell_3^2)^2.\end{aligned} \tag{3.11}$$

Every factor is nonzero for a nondegenerate triangle. For example, $\ell_1 < \ell_2 + \ell_3$ implies

$$\ell_1^2 - \ell_2^2 - 34\ell_2\ell_3 - \ell_3^2 < (\ell_2+\ell_3)^2 - \ell_2^2 - 34\ell_2\ell_3 - \ell_3^2 = -32\ell_2\ell_3 < 0;$$

the remaining assertions follow immediately from the strict triangle inequalities. Hence $\mathcal{E}$ and $\mathcal{S}$ have no common root, a contradiction, which proves (3.7). The polynomials $\mathcal{E}, \mathcal{S}, \mathcal{Q}_i$, the factorization (3.8) and all three resultants are reconstructed in `certificates/edge_fold.py`. □

Proposition 3.4 (Edge disc)**.** *For every nondegenerate triangle and every edge chamber, the set $\{\mathcal{H} \geq 8/9\}$ is a closed disc contained in the open chamber, and $\{\mathcal{H} = 8/9\}$ is its single boundary circle.*

Proof. We first anchor the topology at the equilateral triangle. Put $\mathcal{K}_t = \Phi_t\mathcal{Q} - \Phi\mathcal{Q}_t$ and $\mathcal{K}_s = \Phi_s\mathcal{Q} - \Phi\mathcal{Q}_s$ after setting $\ell_1 = \ell_2 = \ell_3 = 1$; a critical point of $\mathcal{H}$ is exactly a common zero of $\mathcal{K}_s$ and $\mathcal{K}_t$. The complete elimination, rather than a selected-factor check, is

$$\begin{aligned}\operatorname{Res}_t(\mathcal{K}_t, \mathcal{K}_s) = &-3^{15}s^{19}(s-1)^{19}(2s-1)^6(s^2-s+1)^{19}\\ &\cdot (2s^2-2s-1)^2(2s^2-2s+1)^2.\end{aligned}$$

For $0 < s < 1$ this forces $s = 1/2$; substitution gives exactly two critical points in the open square $(0,1)^2$, namely

$$t = \frac{1}{2}, \qquad t = 1 - \frac{\sqrt{3}}{2},$$

with $\mathcal{H}$-values 1 and $-1/3$ respectively. The first is the excenter $(s,t) = (1/2, 1/2)$, at which

$$\mathcal{H} = 1, \qquad \operatorname{Hess}_{(t,s)} \mathcal{H} = \operatorname{diag}(-32, -32/3)$$

in the chart (s,t). Since $\mathcal{H} = 1 - |\Xi|^2 \leq 1$ everywhere, the excenter is the unique global maximum, and it is the only critical point with $\mathcal{H} \geq 8/9$.

By Lemma 3.1, $\mathcal{H} \to 0$ at every compactified end of the chamber, so $\{\mathcal{H} \geq 8/9\}$ is compact and contained in the open chamber. Every connected component of $\{\mathcal{H} \geq 1-\varepsilon\}$ contains a local maximum of $\mathcal{H}$, hence a critical point of value $> 8/9$; there is only one, so by the Morse lemma $\{\mathcal{H} \geq 1-\varepsilon\}$ is a single closed disc for small $\varepsilon > 0$. The band $\{8/9 \leq \mathcal{H} \leq 1-\varepsilon\}$ is compact and contains no critical point, so the vector field

$$X = -\frac{\nabla\mathcal{H}}{\|\nabla\mathcal{H}\|^2}, \qquad d\mathcal{H}(X) = -1,$$

is defined there and its flow exists for the finite time $1 - \varepsilon - 8/9$. It carries $\{\mathcal{H} = 1-\varepsilon\}$ diffeomorphically onto $\{\mathcal{H} = 8/9\}$ and exhibits $\{\mathcal{H} \geq 8/9\}$ as the union of $\{\mathcal{H} \geq 1-\varepsilon\}$ with a collar. Therefore, for the equilateral triangle, $\{\mathcal{H} \geq 8/9\}$ is a closed disc with a single boundary circle.

To continue the *filled* disc to an arbitrary shape, join the equilateral shape to the given one by a path $\tau \mapsto (\xi_0(\tau), \eta_0(\tau))$, $\tau \in [0,1]$, inside the open upper half plane; its image $\mathcal{S}_0$ is compact. By Lemma 3.1 the set

$$\mathcal{Z} = \{(\tau, x) : \mathcal{H}_\tau(x) \geq 8/9\}$$

is compact. Write $\mathcal{Z}_\tau = \{x : (\tau, x) \in \mathcal{Z}\}$ and

$$\mathcal{M} = \partial_{\text{lat}}\mathcal{Z} = \{(\tau, x) : \mathcal{H}_\tau(x) = 8/9\}.$$

Proposition 3.3 gives $d_x\mathcal{H}_\tau \neq 0$ on $\mathcal{M}$. Consequently $\mathcal{Z}$ is a compact manifold with corners whose lateral boundary is $\mathcal{M}$, and both the projection $\pi : \mathcal{Z} \to [0,1]$ and its restriction to $\mathcal{M}$ are proper submersions. On $\mathcal{M}$ the vector field

$$Y = \partial_\tau - \frac{\partial_\tau \mathcal{H}}{\|\nabla_x \mathcal{H}\|^2}\nabla_x \mathcal{H}$$

satisfies $d\pi(Y) = 1$ and $d\mathcal{H}(Y) = 0$; extend its spatial part to a collar neighbourhood of $\mathcal{M}$ in $\mathcal{Z}$ by a partition of unity, keeping $d\pi(Y) = 1$. Compactness makes the flow complete over $[0,1]$, and Ehresmann's fibration theorem for manifolds with boundary [3, Ch. II, §1] gives diffeomorphisms of pairs

$$(\mathcal{Z}_0, \partial\mathcal{Z}_0) \cong (\mathcal{Z}_\tau, \partial\mathcal{Z}_\tau).$$

Thus the equilateral closed disc and its single boundary circle continue to every nondegenerate triangle. □

Proposition 3.5 (The four discs)**.** *$\mathcal{R}$ is the disjoint union of four open discs Ω, one in each admissible sign chamber; each $\overline{\Omega}$ is contained in the open chamber, $\partial\Omega$ is a smooth Jordan curve on which $d|\Xi|^2 \neq 0$, and*

$$\operatorname{wind}_{\partial\Omega} \Xi = -1 \tag{3.12}$$

for the positive boundary orientation.

Proof. By Lemma 2.4, $\overline{\mathcal{R}}$ meets only the four admissible chambers, and its trace on each is $\{\mathcal{H} \geq 8/9\}$; this is a closed disc by Proposition 3.2 in the interior chamber and by Proposition 3.4 in the edge chambers. Regularity of $d|\Xi|^2$ on the boundary holds by Proposition 3.3 in the edge chambers, and in the interior chamber because Ξ is a diffeomorphism there and $|\Xi| = 1/3 \neq 0$.

For (3.12) in the interior chamber, Ξ maps $\partial\Omega$ onto the circle $\{|z| = 1/3\}$ by an orientation-reversing diffeomorphism, so the winding is -1.

For an edge chamber, let $\widehat{C}$ be its compactification as in Lemma 3.1; it is a closed disc. Traverse $\partial\widehat{C}$ positively. Its straight side pieces contribute zero by Lemma 3.1(i). The interior angle of the chamber at a_2 is $\pi - \varphi_2$ and at a_3 is $\pi - \varphi_3$, and the blown-up arcs there are traversed so that $\arg u_i$ *decreases* by the interior angle, exactly as in Proposition 3.2. The chamber subtends the angle φ_1 at infinity, and the arc at infinity is traversed counterclockwise, so $\arg x$ *increases* by φ_1. Since $\Xi = e^{2i\arg(x-a_i)} + o(1)$ at a site and $\Xi = e^{2i\arg x} + o(1)$ at infinity, the total change of $\arg \Xi$ is

$$-2(\pi - \varphi_2) - 2(\pi - \varphi_3) + 2\varphi_1 = -2\pi,$$

using $\varphi_1 + \varphi_2 + \varphi_3 = \pi$. Hence $\operatorname{wind}_{\partial\widehat{C}} \Xi = -1$. Now $\widehat{C} \setminus \Omega$ is a compact annulus on which $|\Xi| \geq 1/3$, so Ξ maps it into $\mathbb{C}^\times$; with the induced annulus orientations the two boundary loops are homotopic there, so

$$\operatorname{wind}_{\partial\widehat{C}} \Xi = \operatorname{wind}^+_{\partial\Omega} \Xi = -1,$$

where $\operatorname{wind}^+_{\partial\Omega}$ denotes winding with the positive boundary orientation of $\overline{\Omega}$. The other edge chambers follow by the cyclic relabelling fixed in Section 1.2, which preserves the plane orientation. □

4 The caustic line family

On a fold circle $\partial\Omega$ choose a local lift

$$3\Xi = e^{i\psi}, \qquad \beta_i = \alpha_i - \psi/2,$$

and define

$$n_i = \frac{\sin\beta_i}{r_i^2}, \qquad \widehat{n}_i = \frac{\cos\beta_i}{r_i^2}, \qquad \nu_i = \lambda_i r_i^3. \tag{4.1}$$

Equivalently $n_i = \Im(e^{-i\psi/2}u_i)/r_i^3$ and $\widehat{n}_i = \Re(e^{-i\psi/2}u_i)/r_i^3$; thus no lift of the individual angle α_i is needed.

Lemma 4.1 (Line family)**.** *Along a fold circle, parametrized regularly by s,*

$$\langle \nu, n\rangle = \langle \nu, \widehat{n}\rangle = 0, \qquad n \times \widehat{n} = -\frac{2S}{(r_1r_2r_3)^3}\,\nu,$$

and $\langle n, \nu_s\rangle = \langle n_s, \nu\rangle = 0$. Consequently $\Theta(x) = [\nu] = [n \times n_s]$ whenever $n \times n_s \neq 0$, and the projective line $\{n \cdot q = 0\}$ is tangent to the caustic $\Theta(\partial\Omega)$ at regular image points and is its limiting tangent at a cusp.

Proof. Barycentric closure gives $\langle \nu, n\rangle = \Im(e^{-i\psi/2}\sum_i \lambda_i u_i) = 0$ and likewise for $\widehat{n}$ with $\Re$ in place of $\Im$. For the cross product, the first component of $n \times \widehat{n}$ is $\sin(\beta_2-\beta_3)/(r_2^2r_3^2) = \sin(\alpha_2-\alpha_3)/(r_2^2r_3^2)$, and $r_2r_3\sin(\alpha_2-\alpha_3) = -\det(u_2,u_3) = -2S\lambda_1$; multiplying numerator and denominator by r_1^3 gives the stated formula, and the other two components follow cyclically.

For the remaining identities, note first that $H_0 = \frac{1}{2}(I + \mathsf{M}(e^{i\psi}))$ on the fold, so $\ker H_0 = \mathbb{R}\, ie^{i\psi/2}$. Put $e_0 = ie^{i\psi/2}$. For any charge variation $\dot q$,

$$\langle e_0, K_x(\dot q)\rangle = -\sum_i \frac{\dot q_i}{r_i^3}\,\Re\big(\overline{ie^{i\psi/2}}\,u_i\big) = -\sum_i \frac{\dot q_i}{r_i^3}\,\Im\big(e^{-i\psi/2}u_i\big) = -\langle \dot q, n\rangle, \tag{4.2}$$

so n is, up to sign, the covector dual to the null direction. Now differentiate the incidence equation $\nabla_P V_{\nu(s)}(x(s)) = 0$ along the fold: by (2.3) this reads $H_0(x)\,x_s + K_x(\nu_s) = 0$. Pairing with e_0 and using that H_0 is symmetric with $H_0e_0 = 0$ gives $\langle e_0, K_x(\nu_s)\rangle = 0$, hence $\langle n, \nu_s\rangle = 0$ by (4.2). Differentiating $\langle n, \nu\rangle = 0$ then gives $\langle n_s, \nu\rangle = 0$.

Thus $n, n_s \in \nu^\perp$, a two-dimensional space, so $n \times n_s$ is parallel to ν whenever it is nonzero. Finally, both ν and ν_s annihilate n, which is the stated tangency. ☐

5 Exact Lami lemma

Every fold circle lies in a strict barycentric chamber, so $r_i > 0$ and $\lambda_i \neq 0$ throughout this section. By a *regular parameter* on a fold circle we mean a parameter s with $x_s \neq 0$.

Proposition 5.1 (Lami regularity)**.** *For every regular parameter s on a fold circle there are smooth functions $\mathfrak{a}, \mathfrak{b}$ such that*

$$n_s = \mathfrak{a}n + \mathfrak{b}\widehat{n}, \qquad \mathfrak{b} \neq 0.$$

Consequently $n \times n_s \neq 0$ everywhere on the fold.

Proof. By Lemma 4.1 the vectors $n, \widehat{n}$ are a basis of the plane $\nu^{\perp}$, which contains n_s; so smooth $\mathfrak{a}, \mathfrak{b}$ exist and the whole content is $\mathfrak{b} \neq 0$. Put

$$\omega = \psi_s, \qquad z_i = e^{-i\psi/2}(x - a_i) = r_i e^{i\beta_i}, \qquad t_i = \tan\beta_i.$$

Step 1: the finite tangent chart. Suppose first that $\cos\beta_i \neq 0$ for all i. On the fold $3\Xi = e^{i\psi}$, so $\sum_i \lambda_i e^{2i\beta_i} = e^{-i\psi}\Xi = 1/3$. Taking real and imaginary parts of this together with $\sum_i \lambda_i = 1$, and writing $\mu_i = \lambda_i/(1+t_i^2)$, gives

$$\sum_i \mu_i = \frac{2}{3}, \qquad \sum_i \mu_i t_i = 0, \qquad \sum_i \mu_i t_i^2 = \frac{1}{3}. \tag{5.1}$$

The t_i are pairwise distinct: equality of two would make the corresponding u_i parallel, putting x on a site line, where $|\Xi| = 1$. Interpolating, one checks that the quadratic $g(t) = \frac{2}{3}t^2 - \frac{2}{3}\mathsf{e}_1 t + \frac{1}{3} + \frac{2}{3}\mathsf{e}_2$, where $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$ are the elementary symmetric functions of t_1, t_2, t_3, satisfies $g(t_i) = \frac{1}{3}(1 + 2t_j t_k)$ and reproduces the three moments (5.1); by uniqueness of the solution of the Vandermonde system,

$$\mu_i = \frac{d_i}{3(t_i - t_j)(t_i - t_k)}, \qquad \lambda_i = \frac{(1+t_i^2)d_i}{3(t_i - t_j)(t_i - t_k)}, \qquad d_i = 1 + 2t_j t_k. \tag{5.2}$$

Here $d_i \neq 0$ because $\lambda_i \neq 0$.

For the radial data put $\eta_i = r_i/\cos\beta_i$, so that $z_i = \eta_i/(1 - it_i)$. The real and imaginary parts of $\sum_i \lambda_i z_i = 0$ are $\sum_i \mu_i \eta_i = 0$ and $\sum_i \mu_i t_i \eta_i = 0$. Hence, with $w_i = \mu_i \eta_i$,

$$(w_1, w_2, w_3) = \varkappa\,(t_2 - t_3,\ t_3 - t_1,\ t_1 - t_2)$$

for some $\varkappa \neq 0$, that is $\eta_i = \varkappa(t_j - t_k)/\mu_i$ for cyclic (i, j, k). By (5.2),

$$\eta_i d_i = 3\varkappa(t_j - t_k)(t_i - t_j)(t_i - t_k) = 3\varkappa\Delta, \qquad \Delta = (t_1 - t_2)(t_1 - t_3)(t_2 - t_3) \neq 0,$$

the last expression being the same for all three cyclic choices. Thus $\eta_i d_i$ is one common nonzero real number $C = 3\varkappa\Delta$, and

$$z_i = \frac{C}{(1 - it_i)d_i}. \tag{5.3}$$

Write $X = e^{-i\psi/2}x_s$, $X/C = p_{\mathrm{L}} + iq_{\mathrm{L}}$ and $\gamma = C_s/C$. The sites are fixed, so $z_{i,s} = X - i\omega z_i/2$, and comparing $z_{i,s}/z_i$ computed from (5.3) with $X/z_i - i\omega/2$ yields

$$\begin{aligned} t_{i,s} &= (1+t_i^2)\{d_i(q_{\mathrm{L}} - p_{\mathrm{L}}t_i) - \omega/2\}, \\ \frac{r_{i,s}}{r_i} &= d_i(p_{\mathrm{L}} + q_{\mathrm{L}}t_i), \\ \gamma - \frac{d_{i,s}}{d_i} - \frac{t_i t_{i,s}}{1+t_i^2} &= d_i(p_{\mathrm{L}} + q_{\mathrm{L}}t_i). \end{aligned} \tag{5.4}$$

Subtracting the last equation for $i = 1$ from those for $i = 2, 3$ eliminates γ and gives a two-by-two linear system for $p_{\mathrm{L}}, q_{\mathrm{L}}$, of determinant

$$-\frac{6(t_1 - t_2)(t_1 - t_3)(t_2 - t_3)\,\mathcal{D}}{d_1 d_2 d_3}, \qquad \mathcal{D} = \sum_i t_i^2 + 4\sum_{i<j} t_i^2 t_j^2 + 12 t_1^2 t_2^2 t_3^2. \tag{5.5}$$

Here $\mathcal{D} > 0$: it is a sum of nonnegative terms which vanishes only if all $t_i = 0$, contrary to their being pairwise distinct. Cramer's rule simplifies to

$$p_{\mathrm{L}} = -\omega\,\frac{2\mathsf{e}_3 - \mathsf{e}_1}{2\mathcal{D}}, \qquad q_{\mathrm{L}} = \omega\,\frac{2\mathsf{e}_2 - 1}{4\mathcal{D}}. \tag{5.6}$$

If $\omega = 0$ then (5.6) gives $X = 0$, hence $x_s = 0$, contradicting regularity of the parameter. Thus $\omega \neq 0$; this conclusion did not presuppose that ψ was a valid parameter.

Step 2: affine compatibility. Differentiating (4.1) and using (5.4),

$$\frac{n_{i,s}}{\widehat{n}_i} = \beta_{i,s} - 2t_i\frac{r_{i,s}}{r_i} = d_i\{(1-2t_i^2)q_{\mathrm{L}} - 3t_i p_{\mathrm{L}}\} - \omega/2. \tag{5.7}$$

We claim that the right-hand side is an *affine* function of t_i whose coefficients do not depend on i; since $n_i = t_i\widehat{n}_i$, this is precisely the assertion $n_s = \mathfrak{a}n + \mathfrak{b}\widehat{n}$. Indeed, $t_jt_k = \mathsf{e}_2 - t_i(\mathsf{e}_1 - t_i)$, so

$$d_i = 1 + 2\mathsf{e}_2 - 2\mathsf{e}_1 t_i + 2t_i^2,$$

and (5.7) becomes a quartic polynomial in t_i with i-independent coefficients. Reduce it modulo the common minimal relation $t^3 = \mathsf{e}_1t^2 - \mathsf{e}_2t + \mathsf{e}_3$, using $t^4 = (\mathsf{e}_1^2 - \mathsf{e}_2)t^2 + (\mathsf{e}_3 - \mathsf{e}_1\mathsf{e}_2)t + \mathsf{e}_1\mathsf{e}_3$. The coefficient of t^2 in the reduced form is

$$(-2(1+2\mathsf{e}_2) + 2)q_{\mathrm{L}} + 4\mathsf{e}_2 q_{\mathrm{L}} = 0$$

identically, the terms in $\mathsf{e}_1 p_{\mathrm{L}}$ and $\mathsf{e}_1^2 q_{\mathrm{L}}$ cancelling as well. Hence the reduction is affine, with

$$\mathfrak{a} = -3p_{\mathrm{L}} - 2(\mathsf{e}_1 + 2\mathsf{e}_3)q_{\mathrm{L}}, \qquad \mathfrak{b} = (1+2\mathsf{e}_2)q_{\mathrm{L}} - 6\mathsf{e}_3 p_{\mathrm{L}} - \frac{\omega}{2}. \tag{5.8}$$

Substituting (5.6) into (5.8) and using $\mathcal{D} = \mathsf{e}_1^2 - 2\mathsf{e}_2 + 4\mathsf{e}_2^2 - 8\mathsf{e}_1\mathsf{e}_3 + 12\mathsf{e}_3^2$ yields the exact identity

$$\boxed{\mathfrak{b} = -\omega\,\frac{P_{\mathrm{L}}}{4\mathcal{D}}} \tag{5.9}$$

$$P_{\mathrm{L}} = 1 + 2\sum_i t_i^2 + 2\big[(t_1t_2 + t_1t_3)^2 + (t_1t_2 + t_2t_3)^2 + (t_1t_3 + t_2t_3)^2\big] \;\geq 1 > 0. \tag{5.10}$$

Equations (5.5), (5.6), the reduction in Step 2, and (5.9)–(5.10) are independently reconstructed over $\mathbb{Z}[t_1, t_2, t_3]$ by `certificates/lami_integer.py` and `certificates/lami_symbolic.py`.

Step 3: the projective boundary of the tangent chart. At most one $\cos\beta_i$ can vanish: two vanishing cosines would make the corresponding u_i parallel, putting x on a site line. Suppose $\cos\beta_1 = 0$, put $\varepsilon_* = \sin\beta_1 \in \{\pm 1\}$, and approach the chart boundary through $|t_1| \to \infty$ with t_2, t_3 bounded. Set

$$D_\infty = 1 + 4t_2^2 + 4t_3^2 + 12t_2^2t_3^2,$$

so that $\mathcal{D}/t_1^2 \to D_\infty$. From (5.6),

$$p_{\mathrm{L}} \sim -\frac{\omega(2t_2t_3 - 1)}{2t_1D_\infty}, \qquad q_{\mathrm{L}} \sim \frac{\omega(t_2 + t_3)}{2t_1D_\infty},$$

while $C = \eta_1 d_1 = r_1(1 + 2t_2t_3)/\cos\beta_1$ gives $C/t_1 \to \varepsilon_* r_1(1 + 2t_2t_3)$. Hence $X = C(p_{\mathrm{L}} + iq_{\mathrm{L}})$ tends to

$$X = \omega\Lambda_*, \qquad \Lambda_* = \frac{\varepsilon_* r_1(1 + 2t_2t_3)}{2D_\infty}\big[-(2t_2t_3 - 1) + i(t_2 + t_3)\big] \neq 0. \tag{5.11}$$

Here $1 + 2t_2t_3 = 3\lambda_1 \neq 0$ in this chart by (5.2), and the squared modulus of the bracket is $1 + (t_2 - t_3)^2 + 4t_2^2t_3^2 > 0$. Since $X \neq 0$, this forces $\omega \neq 0$. Moreover

$$\frac{P_{\mathrm{L}}}{4\mathcal{D}} \longrightarrow \frac{2+4(t_2^2+t_2t_3+t_3^2)}{4D_\infty} > 0. \tag{5.12}$$

Since $z_1 = i\varepsilon_* r_1$, equation (5.11) gives

$$\beta_{1,s} = \Im(X/z_1) - \omega/2 = -\omega\frac{(1+2t_2^2)(1+2t_3^2)}{D_\infty} \neq 0,$$

so the chart boundary is crossed transversely; in particular the set of parameters at which some $\cos\beta_i$ vanishes is discrete. By Lemma 4.1 the pair $n, \widehat{n}$ is a smooth basis of $\nu^\perp$, so $\mathfrak{a}$ and $\mathfrak{b}$ are smooth across such a parameter, and by continuity together with (5.9) and (5.12),

$$\mathfrak{b} = -\omega\,\frac{2+4(t_2^2+t_2t_3+t_3^2)}{4D_\infty} \neq 0$$

there as well. The homogeneous algebra of (5.11) and (5.12) is reconstructed in `certificates/lami_projective.py`.

We have therefore proved globally that $n_s = \mathfrak{a}n + \mathfrak{b}\widehat{n}$ with $\mathfrak{b} \neq 0$, and then $n \times n_s = \mathfrak{b}\, n \times \widehat{n} \neq 0$ by Lemma 4.1. □

5.1 The affine charge chart

Fix the chamber signs $\sigma_i = \operatorname{sign}\lambda_i$, set $D_\sigma = \operatorname{diag}(\sigma_1,\sigma_2,\sigma_3)$, $\mathbf{1} = (1,1,1)$ and $\mathsf{z}_0 = \mathbf{1}/3$. For a projective charge ray with $\langle\sigma, q\rangle \neq 0$ put

$$\mathsf{z}([q]) = \frac{D_\sigma q}{\langle\sigma, q\rangle}, \qquad \mathbf{1}\cdot\mathsf{z} = 1,$$

and use affine coordinates $\mathsf{z} = \mathsf{z}_0 + \mathsf{E}y$ with $\mathsf{E}(y_1,y_2) = (y_1, y_2, -y_1-y_2)$. For every x in the fixed strict sign chamber put $\nu(x) = \vartheta(x)$ and

$$M_\sigma(x) = \langle\sigma, \nu(x)\rangle = \sum_i |\lambda_i(x)| r_i(x)^3 > 0,$$

so that $\mathsf{z}(\Theta(x)) = D_\sigma\nu/M_\sigma$ has all coordinates positive. Define Θ_σ throughout the chamber by $\mathsf{z}(\Theta(x)) = \mathsf{z}_0 + \mathsf{E}\Theta_\sigma(x)$.

On the fold put $m = D_\sigma n$. Since $q = \langle\sigma, q\rangle D_\sigma\mathsf{z}$, the projective line $\{n\cdot q = 0\}$ has affine equation

$$\mathbf{N}\cdot y + c_{\mathrm{aff}} = 0, \qquad \mathbf{N} = \mathsf{E}^{\mathsf{T}} m = (m_1 - m_3, m_2 - m_3), \qquad c_{\mathrm{aff}} = \frac{m_1+m_2+m_3}{3}.$$

Lemma 5.2 (Regularity of the dual line)**.** *For all $a, b \in \mathbb{R}^3$ one has* $\det(\mathsf{E}^{\mathsf{T}}a, \mathsf{E}^{\mathsf{T}}b) = \mathbf{1}\cdot(a\times b)$*. Consequently, on every fold circle,*

$$\det(\mathbf{N}, \mathbf{N}_s) = (\sigma_1\sigma_2\sigma_3)\langle n\times n_s, \sigma\rangle = -(\sigma_1\sigma_2\sigma_3)\frac{2S\,\mathfrak{b}}{(r_1r_2r_3)^3}\,M_\sigma \neq 0. \tag{5.13}$$

Proof. The first identity is a direct expansion of both sides. Applying it to $m = D_\sigma n$ and $m_s = D_\sigma n_s$, and using $(D_\sigma a)\times(D_\sigma b) = \det(D_\sigma)D_\sigma(a\times b)$ for diagonal D_σ, gives the middle expression. The last equality follows from $n\times n_s = \mathfrak{b}\, n\times\widehat{n}$, Lemma 4.1 and $\langle\sigma,\nu\rangle = M_\sigma$; it is nonzero by Proposition 5.1. □

6 Degree one and the support graph

Proposition 6.1 (Support function)**.** *Fix a disc Ω and a positively oriented regular parametrization $x(t)$, $t \in \mathbb{R}$, of $\partial\Omega$ with period T_0. Then the projectivized dual normal $[\mathbf{N}]$ is a diffeomorphism of circles; the normal angle θ is a regular parameter; the function*

$$h(\theta(t)) = -c_{\mathrm{aff}}(t)/|\mathbf{N}(t)|$$

is a well-defined real-analytic function on $\mathbb{R}/2\pi\mathbb{Z}$ with $h(\theta+\pi) = -h(\theta)$; and the affine caustic $\Gamma(t) = \Theta_\sigma(x(t))$ satisfies

$$\Gamma(\theta) = h(\theta)\mathbf{u}(\theta) + h'(\theta)\mathbf{v}(\theta), \qquad \Gamma'(\theta) = \rho(\theta)\mathbf{v}(\theta), \quad \rho = h + h'', \tag{6.1}$$

where $\mathbf{u}(\theta) = (\cos\theta, \sin\theta)$ and $\mathbf{v}(\theta) = (-\sin\theta, \cos\theta)$. Moreover $\Gamma(\theta+\pi) = \Gamma(\theta)$ and $\rho(\theta+\pi) = -\rho(\theta)$.

Proof. For arbitrary x in the strict sign chamber and arbitrary lifted phase ψ, define $n_i(x,\psi) = \Im(e^{-i\psi/2}(x-a_i))/r_i^3$, $m = D_\sigma n$ and $\mathbf{N} = \mathsf{E}^{\mathsf{T}} m$. If $\mathbf{N} = 0$ then m is a multiple of $\mathbf{1}$, so $n = g_0\sigma$; barycentric closure gives $0 = \langle \nu, n\rangle = g_0 M_\sigma$, hence $g_0 = 0$. But $n = 0$ would make all three vectors $e^{-i\psi/2}(x-a_i)$ real and the sites collinear. Thus $\mathbf{N}(x,\psi) \neq 0$ throughout the chamber.

By (3.12) we may choose the phase lift so that $\psi(t+T_0) = \psi(t) - 2\pi$. The sign chamber is convex, its defining barycentric inequalities being affine, so for a fixed x_* in it the straight contraction $x_\tau(t) = (1-\tau)x(t) + \tau x_*$ stays in the chamber; keeping $\psi(t)$ fixed gives a nonvanishing homotopy $[\mathbf{N}(x_\tau(t), \psi(t))]$ of projective loops. At $\tau = 1$, writing $\phi = \psi/2$ and letting $\mathbf{C}_0, \mathbf{S}_0 \in \mathbb{R}^2$ be the cosine and sine coefficient vectors of $\mathbf{N}(x_*, \psi)$, one gets from Lemma 5.2 applied to $\phi = 0$ and $\phi = \pi/2$

$$\mathbf{N} = \mathbf{C}_0\cos\phi + \mathbf{S}_0\sin\phi, \qquad \det(\mathbf{C}_0, \mathbf{S}_0) = (\sigma_1\sigma_2\sigma_3)\frac{2S\,M_\sigma(x_*)}{(r_{1,*}r_{2,*}r_{3,*})^3} \neq 0,$$

where $r_{i,*} = |x_* - a_i|$. Since ϕ decreases by π in one period, the endpoint loop has projective degree $+1$ or -1, and hence so does the original loop. By (5.13), $\theta_t = \det(\mathbf{N}, \mathbf{N}_t)/|\mathbf{N}|^2 \neq 0$ for a local normal-angle lift, so $[\mathbf{N}]$ is a local diffeomorphism of circles; its domain is compact, so it is a proper covering, and absolute degree one makes it a global diffeomorphism. In particular θ is a regular parameter, and since all the data are real analytic, so is $t \mapsto \theta$ and its inverse.

All vector lifts are now taken on the universal cover. The half-angle formula gives $(\mathbf{N}, c_{\mathrm{aff}})(t + T_0) = -(\mathbf{N}, c_{\mathrm{aff}})(t)$. Put $U = \mathbf{N}/|\mathbf{N}|$ and choose $\theta : \mathbb{R} \to \mathbb{R}$ with $U(t) = \mathbf{u}(\theta(t))$. If $\varepsilon_{\mathbf{N}} = \deg[\mathbf{N}] \in \{\pm 1\}$ then $\theta(t+T_0) = \theta(t) + \varepsilon_{\mathbf{N}}\pi$, and for either sign

$$h(\theta+\pi) = -h(\theta), \qquad h(\theta+2\pi) = h(\theta),$$

so h is a real-analytic function on $\mathbb{R}/2\pi\mathbb{Z}$; it is not a function on the projective normal circle, although the envelope constructed below does descend modulo π.

By Lemma 4.1 and the definition of Θ_σ,

$$\mathbf{N}\cdot\Gamma + c_{\mathrm{aff}} = \frac{\langle n, \nu\rangle}{M_\sigma} = 0, \qquad \mathbf{N}_t \cdot \Gamma + (c_{\mathrm{aff}}) * t = \frac{\langle n_t, \nu\rangle}{M * \sigma} = 0,$$

the second identity being obtained by differentiating m only. Because $\det(\mathbf{N}, \mathbf{N}_t) \neq 0$, these two equations uniquely identify the actual caustic point with the envelope of the line family, including at cusp parameters. Subtracting the derivative of the first from the second gives $\mathbf{N}\cdot\Gamma_t = 0$. After unit normalization and reparametrization by θ they become

$$\mathbf{u}\cdot\Gamma = h, \qquad \mathbf{u}\cdot\Gamma' = 0, \qquad \mathbf{v}\cdot\Gamma = h',$$

which is (6.1); indeed $\Gamma' = h'\mathbf{u} + h\mathbf{v} + h''\mathbf{v} - h'\mathbf{u} = (h+h'')\mathbf{v}$. Both factors in the first formula change sign under $\theta \mapsto \theta + \pi$, so Γ is π-periodic and ρ is π-antiperiodic. □

7 Exactly three cusps

Work in the affine charge chart of Section 5.1, introduced after the proof of Proposition 5.1 in Section 5: write

$$q(y) = D_\sigma(\mathsf{z}_0 + \mathsf{E}y), \qquad y = \Theta_\sigma(x),$$

and set $\mathcal{A}(x) = D_y \nabla_P V_{q(y)}(x)$. The matrix $\mathcal{A}$ is independent of y because V_q is linear in q, and it is invertible throughout the strict chamber: if $\mathcal{A}(x)v = 0$ then the charge variation $D_\sigma \mathsf{E}v$ lies in $\ker K_x = \mathbb{R}\nu$, while $\langle \sigma, D_\sigma \mathsf{E}v\rangle = \mathbf{1}\cdot\mathsf{E}v = 0$ and $\langle\sigma,\nu\rangle = M_\sigma(x) > 0$; hence $D_\sigma\mathsf{E}v = 0$ and $v = 0$. Linearity in the charges and vanishing at $y = \Theta_\sigma(x)$ give the exact incidence identity

$$\nabla_P V_{q(y)}(x) = \mathcal{A}(x)\{y - \Theta_\sigma(x)\}. \tag{7.1}$$

Let $H(x)$ denote the planar Hessian of $V_{q(\Theta_\sigma(x))}$ at x with the charge *held fixed* while the spatial second derivative is taken. Since $q(\Theta_\sigma(x)) = \nu(x)/M_\sigma(x)$, Lemma 2.3 gives $H = H_0/M_\sigma$; in particular H is positive definite on Ω and $\det H$ has the sign of $1 - 9|\Xi|^2$. Differentiating (7.1) on the incidence graph gives

$$H + \mathcal{A}\, d\Theta_\sigma = 0, \tag{7.2}$$

so $\ker d\Theta_{\sigma,x} = \ker H(x)$ and $\det d\Theta_\sigma = \det H/\det\mathcal{A}$. By Proposition 3.5 the differential $d|\Xi|^2$ does not vanish on $\partial\Omega$, so the Jacobian determinant of Θ_σ vanishes simply there, and $\partial\Omega$ is exactly the singular curve of Θ_σ in the chamber.

7.1 Classification of the boundary singularities

Fix $x_0 \in \partial\Omega$, put $y_0 = \Theta_\sigma(x_0)$, and let $V = V_{q(y_0)}|_P$ with x_0 translated to the origin; all directional derivatives below are of this fixed-charge function. Choose an orthonormal basis (e, f) with e spanning the null line of $H(x_0)$, and put $\kappa = V_{ff} \neq 0$; thus $V_e = V_f = V_{ee} = V_{ef} = 0$ at the origin.

Lemma 7.1 (Fold criterion)**.** *$e(\det H) = \kappa V_{eee}$ at x_0. Hence the null line is transverse to $\partial\Omega$ exactly when $V_{eee} \neq 0$, and tangent exactly when $V_{eee} = 0$.*

Proof. At x_0 one has $H = \operatorname{diag}(0,\kappa)$ in the basis (e,f), so $\operatorname{adj} H = \operatorname{diag}(\kappa, 0)$ and $e(\det H) = \operatorname{tr}(\operatorname{adj} H \cdot e(H)) = \kappa\, e(H_{ee})$. Now $H_{ee}(x) = \partial_e^2 V_{q(\Theta_\sigma(x))}(x)$ depends on x both through the evaluation point and through the charge; the second contribution is $\partial_e^2 V_{\dot q}$ with $\dot q = D_\sigma \mathsf{E}\, d\Theta_{\sigma,x_0}(e) = 0$, because $e \in \ker d\Theta_{\sigma,x_0}$. Hence $e(H_{ee}) = V_{eee}$. Finally $\partial\Omega = \{\det H = 0\}$ near x_0 and $\det H$ vanishes simply there. □

On the candidate-cusp locus $V_{eee} = 0$. Give the null coordinate ξ weight one and the transverse coordinate η weight two. The Taylor expansion has the form

$$V(\xi e + \eta f) = V(0) + \frac{\kappa}{2}\eta^2 + \frac{V_{eee}}{6}\xi^3 + \frac{V_{eef}}{2}\xi^2\eta + \frac{V_{eeee}}{24}\xi^4 + O_{\mathrm{wt}}(5).$$

Solving $V_f = 0$ gives $\eta(\xi) = -V_{eef}\xi^2/(2\kappa) + O(\xi^3)$; substituting this into V gives

$$V(\xi e + \eta(\xi) f) = V(0) + \frac{V_{eee}}{6}\xi^3 + c_4\xi^4 + O(\xi^5), \qquad c_4 = \frac{1}{24}\left(V_{eeee} - \frac{3V_{eef}^2}{\kappa}\right).$$

Write the vectors from the equilibrium to the sites as $r_i(c_i, s_i)$ with $c_i = \cos\theta_i$ and $s_i = \sin\theta_i$. The sourcewise Coulomb derivatives, obtained from the multipole expansion $|x - a_i|^{-1} = \sum_{n\geq 0} |x|^n r_i^{-n-1} P_n(\cos\gamma_i)$ about x_0, in which γ_i is the angle at x_0 between $x - x_0$ and $a_i - x_0$, are

$$\begin{aligned} V_e &= \sum_i \frac{q_i}{r_i^2} c_i, & V_f &= \sum_i \frac{q_i}{r_i^2} s_i, \\ V_{ee} &= 2\sum_i \frac{q_i}{r_i^3} P_2(c_i), & V_{ef} &= 3\sum_i \frac{q_i}{r_i^3} c_i s_i, \\ V_{eee} &= 6\sum_i \frac{q_i}{r_i^4} P_3(c_i), & V_{eef} &= 3\sum_i \frac{q_i}{r_i^4} s_i(5c_i^2 - 1), \\ V_{eeee} &= 24\sum_i \frac{q_i}{r_i^5} P_4(c_i), \end{aligned}$$

where P_j is the jth Legendre polynomial. These formulas are checked term by term in `certificates/coulomb_derivatives.py`.

For cyclic (i, j, k) put

$$\mathfrak{d}_i = \sin(\theta_k - \theta_j), \qquad \mathfrak{D} = \mathfrak{d}_1\mathfrak{d}_2\mathfrak{d}_3, \qquad \mathfrak{p}_i = 2c_j c_k + s_j s_k.$$

The equilibrium equations give $q_i/r_i^2 = \gamma_* \mathfrak{d}_i$ with $\gamma_* \neq 0$. Since every $q_i \neq 0$, all $\mathfrak{d}_i \neq 0$, so the three directions $e^{2i\theta_i}$ are distinct. The two null-Hessian equations $V_{ee} = V_{ef} = 0$ read

$$\sum_i \frac{\mathfrak{d}_i P_2(c_i)}{r_i} = 0, \qquad \sum_i \frac{\mathfrak{d}_i c_i s_i}{r_i} = 0.$$

They have rank two. Otherwise some $(\alpha, \beta) \neq (0, 0)$ would satisfy $\alpha P_2(c_i) + \beta c_i s_i = 0$ for all i; in the coordinates $(X_i, Y_i) = (\cos 2\theta_i, \sin 2\theta_i)$, and using $P_2(c) = (1 + 3X)/4$ and $cs = Y/2$, this is the nonzero affine line $\alpha(1 + 3X) + 2\beta Y = 0$ through three distinct points of the unit circle, which is impossible. Direct substitution gives the exact identities

$$\sum_i \mathfrak{d}_i P_2(c_i)\mathfrak{p}_i = 0, \qquad \sum_i \mathfrak{d}_i c_i s_i \mathfrak{p}_i = 0, \qquad \sum_i \mathfrak{d}_i \mathfrak{p}_i = -3\mathfrak{D}.$$

The third shows that $(\mathfrak{p}_i)$ is nonzero and the first two show that it lies in the common kernel; rank two makes that kernel its span. Hence

$$\frac{1}{r_i} = \tau_0\, \mathfrak{p}_i, \qquad \tau_0 \neq 0,$$

and in particular every $\mathfrak{p}_i \neq 0$. Define

$$\begin{aligned} \widehat{E} &= \sum_i \mathfrak{d}_i \mathfrak{p}_i^2 P_3(c_i), & \widehat{U} &= \sum_i \mathfrak{d}_i \mathfrak{p}_i^2 s_i(5c_i^2 - 1), \\ \widehat{W} &= \sum_i \mathfrak{d}_i \mathfrak{p}_i^3 P_4(c_i), & \mathcal{I} &= 3\widehat{U}^2 + 8\mathfrak{D}\widehat{W}. \end{aligned}$$

Using $\sum_i \mathfrak{d}_i \mathfrak{p}_i = -3\mathfrak{D}$ and harmonicity (which gives $\kappa = V_{ff} = \sum_i q_i/r_i^3$ when $V_{ee} = 0$), the derivative formulas give

$$\kappa = -3\gamma_*\tau_0\mathfrak{D}, \quad V_{eee} = 6\gamma_*\tau_0^2\widehat{E},$$
$$V_{eef} = 3\gamma_*\tau_0^2\widehat{U}, \quad V_{eeee} = 24\gamma_*\tau_0^3\widehat{W}.$$

With $\varpi = \gamma_*\tau_0^2 \neq 0$, substitution yields

$$\kappa c_4 = -\frac{3}{8}\varpi^2\,\mathcal{I}. \tag{7.3}$$

Lemma 7.2 (Strict positivity)**.** *At every candidate cusp, $\kappa c_4 < 0$.*

Proof. Set

$$\begin{aligned}
\Upsilon_0 &= c_1c_2c_3, \qquad \Upsilon_3 = s_1s_2s_3,\\
\Upsilon_1 &= s_1c_2c_3 + c_1s_2c_3 + c_1c_2s_3,\\
\mathcal{C} &= s_1s_2c_3 + s_1c_2s_3 + c_1s_2s_3 - 2\Upsilon_0,\\
\mathcal{F} &= 12\Upsilon_1^2 - 14\Upsilon_1\Upsilon_3 + 3\Upsilon_3^2 + 8\Upsilon_0^2.
\end{aligned}$$

Exact Laurent-polynomial expansion gives

$$\widehat{E} = -\mathfrak{D}\,\mathcal{C}, \qquad \widehat{U} = 4\mathfrak{D}\,\Upsilon_1, \qquad \mathcal{I} = \mathfrak{D}^2\{2\mathcal{F} - 8\mathcal{C}(\mathcal{C} + 7\Upsilon_0)\}.$$

The candidate-cusp condition is $\widehat{E} = 0$; since $\mathfrak{D} \neq 0$ this means $\mathcal{C} = 0$, and then $\mathcal{I} = 2\mathfrak{D}^2\mathcal{F}$.

If $\Upsilon_0 \neq 0$, put $t_i = \tan\theta_i$ and let $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$ be their elementary symmetric functions. Then $\mathcal{C} = 0$ reads $\mathsf{e}_2 = 2$, and

$$\begin{aligned}
\frac{\mathcal{F}}{\Upsilon_0^2} &= 12\mathsf{e}_1^2 - 14\mathsf{e}_1\mathsf{e}_3 + 3\mathsf{e}_3^2 + 8\\
&= \frac{184}{3} + 12(\mathsf{e}_1^2 - 3\mathsf{e}_2) + \frac{14}{3}(\mathsf{e}_2^2 - 3\mathsf{e}_1\mathsf{e}_3) + 3\mathsf{e}_3^2 > 0,
\end{aligned}$$

because

$$\begin{aligned}
\mathsf{e}_1^2 - 3\mathsf{e}_2 &= \tfrac{1}{2}\textstyle\sum_{i<j}(t_i - t_j)^2,\\
\mathsf{e}_2^2 - 3\mathsf{e}_1\mathsf{e}_3 &= \tfrac{1}{2}\big[(t_1t_2 - t_1t_3)^2 + (t_1t_2 - t_2t_3)^2 + (t_1t_3 - t_2t_3)^2\big].
\end{aligned}$$

If $\Upsilon_0 = 0$, at most one cosine vanishes because the directions $e^{2i\theta_i}$ are distinct; after relabelling $c_3 = 0$, so $s_3 = \pm 1$ and $\mathcal{C} = s_3\sin(\theta_1 + \theta_2)$. Thus $\mathcal{C} = 0$ forces $\theta_2 \equiv -\theta_1 \pmod{\pi}$, whence $c_1c_2 = \pm c_1^2$ and $s_1s_2 = \mp s_1^2$ with matching signs, and

$$\mathcal{F} = 12c_1^2c_2^2 - 14c_1c_2s_1s_2 + 3s_1^2s_2^2 = 12c_1^4 + 14c_1^2s_1^2 + 3s_1^4 = c_1^4 + 8c_1^2 + 3 > 0.$$

In both cases $\mathcal{I} = 2\mathfrak{D}^2\mathcal{F} > 0$, so (7.3) gives $\kappa c_4 < 0$. □

Proposition 7.3 (Boundary singularities)**.** *Every singularity of Θ_σ on $\partial\Omega$ is either a fold ($V_{eee} \neq 0$) or an ordinary Whitney cusp ($V_{eee} = 0$). In Arnold's notation [1], the reduced potential at a cusp has type A_3; types A_k with $k \geq 4$ do not occur.*

Proof. Lemma 7.1 identifies the fold points. Let x_0 be a candidate cusp, so $V_{eee} = 0$ and, by Lemma 7.2 and $\kappa \neq 0$, also $c_4 \neq 0$; the reduced potential is then $V(0) + c_4\xi^4 + O(\xi^5)$, of type A_3.

It remains to verify versality. Let $\eta(\xi)$ be the solution of $V_f = 0$ used above and set $\mathfrak{c}(\xi) = \xi e + \eta(\xi)f$. If $\mathfrak{g}(\xi) = V(\mathfrak{c}(\xi))$ then $\mathfrak{g}'(\xi) = V_e(\mathfrak{c}(\xi))$, so

$$\nabla_P V_{q(y_0)}(\mathfrak{c}(\xi)) = (4c_4\xi^3 + O(\xi^4),\, 0)$$

in the (e, f) components. Putting $\mathcal{A}_0 = \mathcal{A}(x_0)$, (7.1) yields

$$\Theta_\sigma(\mathfrak{c}(\xi)) - y_0 = -4c_4\xi^3 \mathcal{A}_0^{-1}(1,0) + O(\xi^4). \tag{7.4}$$

By (7.2), $\operatorname{im} d\Theta_{\sigma,x_0} = \mathcal{A}_0^{-1} \operatorname{im} H(x_0) = \mathbb{R}\, \mathcal{A}_0^{-1}(0,1)$, so the cubic vector in (7.4) is transverse to the rank-one image of $d\Theta_{\sigma,x_0}$.

We spell this out in rank coordinates. Since $d\Theta_{\sigma,x_0}$ has rank one, choose target coordinates whose first axis is $\operatorname{im} d\Theta_{\sigma,x_0}$ and then source coordinates (s,t) in which the map germ becomes $\Theta_\sigma - y_0 = (s, \mathsf{h}(s,t))$, with the kernel equal to the t-axis; the Jacobian is h_t, and $d\mathsf{h}(0) = 0$. Kernel tangency gives $\mathsf{h}_{tt}(0) = 0$, while simple vanishing of the Jacobian gives $\mathsf{h}_{st}(0) \neq 0$. In these coordinates $t(\mathfrak{c}(\xi)) = \chi\xi + O(\xi^2)$ with $\chi \neq 0$, because $\mathfrak{c}'(0) = e$ spans the kernel, and $s(\mathfrak{c}(\xi)) = O(\xi^3)$, because s is the first component of $\Theta_\sigma - y_0$ and (7.4) applies. Since $d\mathsf{h}(0) = 0$ and $\mathsf{h}_{tt}(0) = 0$, the transverse component in (7.4) has cubic coefficient $\chi^3\mathsf{h}_{ttt}(0)/6$; hence $\mathsf{h}_{ttt}(0) \neq 0$. Whitney's normal form for such a germ [15], [6, Ch. VI, Thm. 2.2], puts it in orientation-preserving source and target coordinates into the ordinary cusp form

$$\mathsf{W}(s,t) = (s,\, \mathsf{a}t^3 + \mathsf{b}st), \qquad \mathsf{ab} \neq 0. \tag{7.5}$$

The Laurent identities of Lemma 7.2 and both positivity charts are reconstructed in `certificates/cusp_quartic.py`. ☐

7.2 The cusp count

Proposition 7.4 (Exactly three cusps). *Every fold circle carries exactly three cusps, and ρ in (6.1) has exactly three simple zeros, hence exactly three sign changes, in one projective period.*

Proof. Orient $\mathbb{RP}^1$ by increasing angle modulo π, parametrize $\partial\Omega$ positively, and define

$$\mathsf{T}, \mathsf{K} : \partial\Omega \longrightarrow \mathbb{RP}^1, \qquad \mathsf{T}(x) = T_x\partial\Omega, \qquad \mathsf{K}(x) = \ker d\Theta_{\sigma,x}.$$

At an intersection choose local angle lifts, put $\delta = \angle\mathsf{T} - \angle\mathsf{K}$, and set $\epsilon = \operatorname{sign}\delta'$. The turning-number theorem for the positively oriented Jordan curve $\partial\Omega$ gives $\deg \mathsf{T} = 2$. By (3.12) a lift of ψ decreases by 2π around the boundary, and by Lemma 2.3 together with (7.2) one has $\mathsf{K}(x) = \ker H(x) = \mathbb{R}\, ie^{i\psi(x)/2}$, of degree -1. Cusps are exactly the zeros of δ, and they are transverse by the computation below. The signs ϵ are the local coincidence signs of the two circle maps, so their sum is the difference of the two degrees:

$$\sum_{\text{cusps}} \epsilon = \deg \mathsf{T} - \deg \mathsf{K} = 2 - (-1) = 3. \tag{7.6}$$

It remains to show that no signs cancel. Orient the target so that $\det d\Theta_\sigma > 0$ in Ω. Since H is positive definite in Ω, (7.2) gives $\det \mathcal{A} = \det H / \det d\Theta_\sigma > 0$ in Ω; invertibility and continuity therefore give $\det \mathcal{A} > 0$ on $\partial\Omega$ as well. At a cusp x_0 with $y_0 = \Theta_\sigma(x_0)$, (7.1) reads

$$\nabla_P V_{q(y_0)}(x) = -\mathcal{A}(x)\{\Theta_\sigma(x) - y_0\}.$$

The normal form (7.5) shows that x_0 is an isolated preimage of y_0, so both sides have a well-defined local degree. Both $\mathcal{A}(x)$ and $-I$ lie in $GL^+(2,\mathbb{R})$, which is connected, so after shrinking an isolating neighbourhood of x_0 the matrix field $-\mathcal{A}(x)$ is homotopic to the identity through $GL^+(2,\mathbb{R})$-valued fields; multiplication by this homotopy creates no boundary zero, whence

$$\deg_{x_0}(\Theta_\sigma - y_0) = \deg_{x_0} \nabla_P V_{q(y_0)}.$$

By the splitting lemma the germ of $V_{q(y_0)}$ at x_0 is right equivalent to $c_4\xi^4 + \frac{\kappa}{2}\eta^2$, whose gradient $(4c_4\xi^3, \kappa\eta)$ has local degree $\operatorname{sign}(\kappa c_4) = -1$ by Lemma 7.2.

In the orientation-preserving coordinates of (7.5) the Jacobian is $J_{\mathsf{W}} = 3\mathsf{a}t^2 + \mathsf{b}s$, which is positive on the side corresponding to Ω, and the local degree of W is $\operatorname{sign}\mathsf{a}$; hence $\mathsf{a} < 0$. The singular curve is $s = -3\mathsf{a}t^2/\mathsf{b}$, and its positive-boundary parametrization is

$$(s,t) = \Big(-\frac{3\mathsf{a}}{\mathsf{b}}\zeta^2,\ -\operatorname{sign}(\mathsf{b})\,\zeta\Big),$$

along which the kernel of $d\mathsf{W}$ is the constant t-axis. A direct computation gives $\delta'(0) = -6\mathsf{a}/|\mathsf{b}| \neq 0$, so the zeros of δ are transverse and $\epsilon = -\operatorname{sign}\mathsf{a} = +1$ at every cusp. With (7.6) this gives exactly three cusps.

Finally, the normal angle is a regular parameter by Proposition 6.1, and a cusp is exactly a zero of ρ in (6.1), since $\Gamma' = d\Theta_\sigma(x_t)$ vanishes precisely when the tangent lies in the kernel. At such a zero $\Gamma'' = \rho'\mathbf{v}$. Restricting (7.5) to its singular curve gives $\mathsf{W} = (-3\mathsf{a}\zeta^2/\mathsf{b},\ 2\mathsf{a}\operatorname{sign}(\mathsf{b})\zeta^3)$, whose second derivative at $\zeta = 0$ is nonzero; hence $\rho' \neq 0$. Thus the three cusp parameters are exactly three simple zeros, equivalently three sign changes, of ρ in one projective period. Figure 2 shows a caustic with its three cusps. □

8 Three-sign-change envelopes are embedded

For a continuous real function ρ on an open interval I, define its alternation number by

$$\operatorname{alt}_I(\rho) = \sup\left\{ m \in \mathbb{Z}_{\geq 0} : \begin{array}{c} x_0 < \cdots < x_m \text{ in } I, \\ \rho(x_j)\rho(x_{j+1}) < 0 \quad (0 \leq j < m) \end{array} \right\}.$$

The value $m = 0$ is always attained, so the supremum is over a nonempty set. Zeros are omitted in this definition; thus a tangential zero contributes no alternation, while a genuine crossing contributes one. The value $+\infty$ is allowed.

Lemma 8.1. *Let $h \in C^2(\mathbb{R}/2\pi\mathbb{Z})$ satisfy $h(\theta+\pi) = -h(\theta)$, put $\rho = h + h''$, and define Γ by (6.1). Suppose that ρ is not identically zero on any nonempty open interval and that*

$$\sup_{t\in\mathbb{R}} \operatorname{alt}_{(t,t+\pi)}(\rho) \leq 3.$$

Then $\Gamma : \mathbb{R}/\pi\mathbb{Z} \to \mathbb{R}^2$ is injective.

Proof. Identify $\mathbb{R}^2$ with $\mathbb{C}$. Differentiating (6.1) gives $\Gamma'(s) = i\rho(s)e^{is}$, hence

$$\Gamma(b) - \Gamma(a) = i\int_a^b \rho(s)e^{is}\,ds. \tag{8.1}$$

Claim. If $I = (a,b)$ has length less than π and $\operatorname{alt}_I(\rho) \leq 1$, then the integral in (8.1) does not vanish.

If there is no alternation, all nonzero values of ρ have one sign, while $\cos(s-(a+b)/2) > 0$ on I because $|s-(a+b)/2| < \pi/2$. Therefore

$$\Re\left(e^{-i(a+b)/2}\int_a^b \rho(s)e^{is}\,ds\right) = \int_a^b \rho(s)\cos\left(s-\tfrac{a+b}{2}\right)ds$$

has a strict sign. If there is exactly one alternation, we first produce a cut. Let $\mathcal{P}_+ = \{\rho > 0\} \cap I$ and $\mathcal{P}_- = \{\rho < 0\} \cap I$. If neither $\sup \mathcal{P}_+ \le \inf \mathcal{P}_-$ nor $\sup \mathcal{P}_- \le \inf \mathcal{P}_+$ held, there would exist $p \in \mathcal{P}_+, n \in \mathcal{P}_-$ with $p < n$ and $p' \in \mathcal{P}_+, n' \in \mathcal{P}_-$ with $n' < p'$; comparing n' with p produces in either case a three-term alternating sample, so $\mathrm{alt}_I(\rho) \ge 2$. Hence a cut $c \in [a, b]$ exists, and after reversing the overall sign if necessary ρ is nonnegative on (a, c) and nonpositive on (c, b). Since $b - a < \pi$, the weight $-\sin(s - c)$ has exactly the same sign pattern, so

$$\Re\left(e^{-i(c-\pi/2)}\int_a^b \rho(s)e^{is}\,ds\right) = \int_a^b -\rho(s)\sin(s-c)\,ds$$

also has a strict sign. Strictness in both cases follows from continuity and the hypothesis that ρ vanishes on no open interval. This proves the claim.

Suppose now that $\Gamma(a) = \Gamma(b)$ for distinct projective parameters, and choose lifts with $0 < b - a < \pi$. Since $\Gamma(a + \pi) = \Gamma(a)$, equation (8.1) vanishes on both (a, b) and $(b, a + \pi)$, and both intervals have length less than π. By the claim each carries at least two alternations, so each contains three interior sample points with alternating signs. Concatenating the two sample sequences gives six points of $(a, a + \pi)$; if the two signs at the join agree, drop one of them. The resulting sequence has at least four alternations in $(a, a + \pi)$, which contradicts the hypothesis. □

Corollary 8.2 (Jordan caustics)**.** *For each of the four discs Ω, the caustic $\Gamma_\Omega := \Theta_\sigma(\partial\Omega)$ is a Jordan curve, and the three cusp images are distinct.*

Proof. By Proposition 6.1 all line-family data are real analytic and the normal angle is regular, so h is real analytic and ρ vanishes on no open interval. By Proposition 7.4, ρ has exactly three simple zeros in a projective period, so every open interval of length π carries at most three alternations. Lemma 8.1 therefore applies and Γ is injective on $\mathbb{R}/\pi\mathbb{Z}$; being continuous on a compact space, it is a homeomorphism onto its image. The projective caustic $\Theta(\partial\Omega)$ is the image of Γ_Ω under the inverse affine chart and is therefore a Jordan curve as well. The term *Jordan* is meant topologically: Γ is a continuous injective parametrization of its image, but it is not an immersion at its three ordinary cusps. □

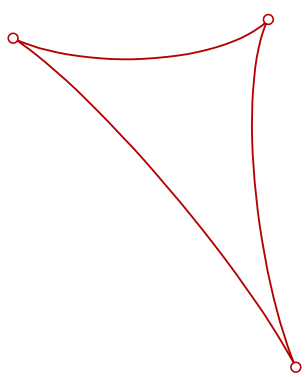

Figure 2: The caustic $\Gamma_\Omega = \Theta_\sigma(\partial\Omega)$ of the interior disc of Figure 1, drawn in the affine charge chart of Section 5.1 and centred at its barycentre. By Proposition 7.4 it has exactly three ordinary cusps (marked), all of the same coincidence sign, and by Corollary 8.2 it is a Jordan curve.

9 Fibres and the final count

All gradient indices and windings in this section are two-dimensional: they are taken in the site plane P in the noncollinear case and in a fixed meridian plane in the collinear case. We write $\operatorname{ind}_x \nabla V$ for the local degree of a planar gradient at an isolated zero, and

$$\mathfrak{N} = \#\operatorname{Crit}(V_q).$$

Proposition 9.1 (Fibre alternatives). *Let the sites be noncollinear and put*

$$m = \#(\Theta^{-1}([q]) \cap \mathcal{R}), \qquad d = \#(\Theta^{-1}([q]) \cap \partial\mathcal{R}).$$

If $[q]$ lies on no caustic, then $m \le 1$ and $d = 0$. If $[q]$ lies on a caustic, then $m = 0$ and $d = 1$, and the corresponding degenerate equilibrium has planar gradient index $\iota \in \{0, -1\}$, namely 0 at a boundary fold and -1 at a boundary cusp.

Proof. Fix a disc Ω and work in the affine target chart of its strict charge-sign chamber. For a target point y off Γ_Ω, every preimage in Ω is regular and all local degrees of $\Theta_\sigma|_\Omega$ have the same sign, so

$$\#(\Theta_\sigma^{-1}(y) \cap \Omega) = |\deg(\Theta_\sigma, \Omega, y)| = |\operatorname{wind}(\Gamma_\Omega, y)| \le 1, \tag{9.1}$$

the last inequality in (9.1) holding because Γ_Ω is a Jordan curve (Corollary 8.2). Now let $y \in \Gamma_\Omega$. Boundary injectivity gives exactly one boundary preimage. There is no interior preimage: if there were, its local inverse would persist for target points on both sides of the Jordan curve, and choosing a nearby point on the side with winding zero would give a nonempty fibre whose local degrees all have one sign, contradicting degree zero.

By Lemma 2.4 a fibre of Θ meets at most one of the four closed admissible chambers, hence at most one $\overline{\Omega}$. Combining this with the two paragraphs above gives the stated alternatives; in particular $[q]$ cannot lie on two caustics. Finally, Proposition 7.3 and the splitting normal forms of Section 7 give the planar gradient index at a boundary point: at a fold the reduced potential is $\frac{V_{eee}}{6}\xi^3 + \frac{\kappa}{2}\eta^2$ with $V_{eee} \neq 0$, whose gradient has local degree 0, and at a cusp it is $c_4\xi^4 + \frac{\kappa}{2}\eta^2$, of local degree $\operatorname{sign}(\kappa c_4) = -1$. □

Remark 9.2. Proposition 9.1 determines ι exactly, but only the bound $\iota \le 1$ is used below. That weaker bound also follows, independently of Section 7, from Khimshiashvili's formula [7], [1, Ch. I, §3]: if $f : (\mathbb{R}^2, 0) \to (\mathbb{R}, 0)$ is real analytic with an isolated critical point and $0 < \delta_{\mathrm{M}} \ll \varepsilon_{\mathrm{M}} \ll 1$ are Milnor data, then

$$\operatorname{ind}_0 \nabla f = 1 - \chi(F_-) \le 1, \qquad F_- = f^{-1}(f(0) - \delta_{\mathrm{M}}) \cap \overline{B}_{\varepsilon_{\mathrm{M}}}(0),$$

because F_- is a compact one-manifold, possibly empty or with boundary, and therefore has nonnegative Euler characteristic.

Proof of Theorem 1, noncollinear case. By Lemma 2.1 we may work entirely in P, and by the finite-critical-set hypothesis the zero set of $\nabla_P V_q$ consists of finitely many isolated points. No equilibrium lies on a site line, since $\lambda_k = 0$ would force $q_k = 0$ by (2.1); so every equilibrium lies in a strict chamber, and by Lemma 2.3 an equilibrium outside $\overline{\mathcal{R}}$ is a regular saddle of planar index -1.

Choose pairwise disjoint isolating circles and apply additivity of degree on the punctured plane. Let w_∞ be the winding of $\nabla_P V_q$ on a sufficiently large positively oriented circle. Near a_i the field is $-q_i\varepsilon^{-2}e^{i\theta} + O(1)$ on the circle of radius ε, so each site contributes winding $+1$ counterclockwise; the induced boundary orientation of a puncture is clockwise, so these three windings are subtracted:

$$\sum_{x\in\mathrm{Crit}(V_q)} \mathrm{ind}_x\, \nabla_P V_q = w_\infty - 3. \tag{9.2}$$

Let $\mathfrak{s}$ be the number of equilibria outside $\overline{\mathcal{R}}$. By Proposition 9.1 and (9.2) there are two cases. If $d = 0$ then $m \le 1$ and $\mathfrak{N} = m + \mathfrak{s}$ with $m - \mathfrak{s} = w_\infty - 3$, so

$$\mathfrak{N} = 2m + 3 - w_\infty \le 5 - w_\infty. \tag{9.3}$$

If $d = 1$ then $m = 0$ and $\mathfrak{N} = \mathfrak{s} + 1$ with $\iota - \mathfrak{s} = w_\infty - 3$, so

$$\mathfrak{N} = \iota + 4 - w_\infty \le 5 - w_\infty. \tag{9.4}$$

It remains to compute the outer winding in (9.2). Put $\mathsf{Q} = \sum_i q_i$ and $\mathbf{p} = \sum_i q_i a_i$. Uniformly for $x = R\omega$ with $|\omega| = 1$,

$$\nabla_P V_q(R\omega) = -\frac{\mathsf{Q}}{R^2}\omega + \frac{\mathbf{p} - 3(\mathbf{p}\cdot\omega)\omega}{R^3} + O(R^{-4}). \tag{9.5}$$

If $\mathsf{Q} \ne 0$, the leading field in (9.5) is radial and $w_\infty = 1$; then (9.3)–(9.4) give $\mathfrak{N} \le 4$. If $\mathsf{Q} = 0$, affine independence implies $\mathbf{p} \ne 0$, for otherwise (q_1, q_2, q_3) would be a nonzero affine dependence among the sites. Rotate and scale so that $\mathbf{p}$ is the positive real unit vector. For $\omega = e^{i\theta}$,

$$\mathbf{p} - 3(\mathbf{p}\cdot\omega)\omega = -\tfrac{1}{2}(1 + 3e^{2i\theta}),$$

which never vanishes and winds twice. Thus $w_\infty = 2$ and $\mathfrak{N} \le 3$. □

Remark 9.3 (The zero-sum case). Let the sites be noncollinear with $\mathsf{Q} = 0$, so $w_\infty = 2$ and (9.3), (9.4) read $\mathfrak{N} = 2m + 1$ and $\mathfrak{N} = \iota + 2$ respectively. Hence

$$\mathfrak{N} = 3 \quad\Longleftrightarrow\quad m = 1 \ \text{ and } \ d = 0.$$

Since $q = \varsigma\,\vartheta(x)$ at every equilibrium, it is convenient to put

$$\mathfrak{m}_k(x) = \sum_i \lambda_i(x) r_i(x)^k,$$

so that for the canonical representative $\mathfrak{m}_0 = \sum_i q_i/r_i^3 = 1$, $\mathfrak{m}_2 = V_{\vartheta(x)}(x) = R_c^2 - |x - O|^2$ and $\mathfrak{m}_3 = \mathsf{Q}$. Thus $\mathsf{Q} = 0$ holds exactly on the locus $\{\mathfrak{m}_3 = 0\}$, and

$$\text{the constant 3 is attained} \iff \mathfrak{m}_3 \text{ vanishes somewhere on one of the four discs } \Omega. \tag{9.6}$$

On the interior disc every $\lambda_i > 0$, so $\mathfrak{m}_3 > 0$ there and only the three edge discs can contribute. On an edge disc the following hold throughout $\overline{\Omega}$:

(i) $\mathfrak{m}_0 = 1$;

(ii) $\mathfrak{m}_1 > 0$: the relation $\sum_i \lambda_i u_i = 0$ exhibits $|\lambda_1| r_1, \lambda_2 r_2, \lambda_3 r_3$ as the side lengths of a triangle, nondegenerate because $|\Xi| \le 1/3 < 1$ keeps x off the site lines, so $|\lambda_1| r_1 < \lambda_2 r_2 + \lambda_3 r_3$;

(iii) $\mathfrak{m}_2 < 0$, by Lemma 2.4.

These three do not by themselves force $\mathfrak{m}_3 < 0$. Normalizing $r_1 = 1$ and writing $a = r_2$, $b = r_3$, the system $\mathfrak{m}_0 = 1$, $\mathfrak{m}_1 = \mathfrak{m}_2 = 0$ is solved by

$$\lambda_1 = -\frac{ab}{(1-a)(b-1)}, \qquad \lambda_2 = \frac{b}{(1-a)(b-a)}, \qquad \lambda_3 = \frac{a}{(b-a)(b-1)},$$

which has the required signs for $0 < a < 1 < b$, and there $\mathfrak{m}_3 = ab > 0$. Of course $\mathfrak{m}_2 = 0$ forces $|\Xi| = 1$, so the constraint $|\Xi| \le 1/3$ is essential.

A Laplace representation narrows the possibilities. For the canonical representative put $\phi(s) = \sum_i q_i e^{-r_i^2 s}$; then

$$\int_0^\infty s^{1/2}\phi(s)\,ds = \frac{\sqrt{\pi}}{2}\mathfrak{m}_0 > 0, \qquad \int_0^\infty s^{-1/2}\phi(s)\,ds = \sqrt{\pi}\,\mathfrak{m}_2 < 0, \qquad \phi(0^+) = \mathfrak{m}_3.$$

By Descartes' rule for exponential sums, ϕ has at most two sign changes on $(0,\infty)$. If it has none then $\mathfrak{m}_0$ and $\mathfrak{m}_2$ share a sign, which is false. If its sign pattern begins with $-$, then $\mathfrak{m}_3 = \phi(0^+) \le 0$. If the pattern is $(+,-)$ with the change at s_*, then $s \mapsto s$ is increasing while $s^{-1/2}\phi$ is positive before s_* and negative after, so $\int_0^\infty s^{1/2}\phi < s_* \int_0^\infty s^{-1/2}\phi < 0$, contradicting $\mathfrak{m}_0 = 1$. Hence the only pattern compatible with $\mathfrak{m}_3 \ge 0$ is $(+,-,+)$, and that forces the odd-sign charge to sit at the middle of the three distances r_i.

We have not been able to exclude this last case in closed form, but it can be searched exhaustively, because (9.6) is a four-parameter semialgebraic question. Indeed $\sum_i \lambda_i u_i = 0$ determines the directions u_i from (λ, r) up to a rotation, hence determines $|\Xi|$; conversely every (λ, r) with $\sum_i \lambda_i = 1$ and $|\lambda_i| r_i$ obeying the strict triangle inequality is realized by an actual configuration, on taking $x = 0$ and $a_i = -u_i$. Explicitly, with $B_i = \lambda_i^2 r_i^2$ and $\mathcal{J} = 2\sum_{i<j} B_i B_j - \sum_i B_i^2$,

$$1 - |\Xi|^2 = \frac{\mathcal{J}\,\mathfrak{m}_2}{\lambda_1\lambda_2\lambda_3\, r_1^2 r_2^2 r_3^2},$$

so after normalizing $r_1 = 1$ the feasible set is cut out in $(\lambda_1, \lambda_2, r_2, r_3)$ by $\mathcal{J} > 0$ and $1 - |\Xi|^2 \ge 8/9$. Extensive search of that set — global sampling, and constrained local optimization from many starts, including a dedicated sweep of the $(+,-,+)$ regime — produced no point with $\mathfrak{m}_3 \ge 0$. Accordingly we expect the following, which would also match the collinear zero-sum case, where $M_0 = 0$ forces $\ell \ge 1$ and hence $\mathfrak{N} \le 2 - \ell \le 1$.

Conjecture. *If the sites are noncollinear and $\sum_i q_i = 0$, then $\mathfrak{m}_3 < 0$ on each of the three edge discs. Consequently $m = d = 0$, and V_q has exactly one equilibrium, a saddle.*

9.1 Collinear sites

Suppose finally that the sites are collinear, at $(h_i, 0, 0)$ on the first coordinate axis. An off-axis equilibrium would generate a full critical circle under rotation about that axis, so finiteness forces every equilibrium onto the axis.

Fix a meridian plane P_{m} containing the axis, and identify its oriented coordinates with $x_1 + ix_2$, so that the axis is the real axis. Since V_q is invariant under the reflection fixing P_{m}, its normal derivative vanishes identically on P_{m}; hence the zeros of $\nabla_{P_{\mathrm{m}}} V_q$ in P_{m} are exactly $\mathrm{Crit}(V_q) \cap P_{\mathrm{m}}$, which by the previous paragraph is all of $\mathrm{Crit}(V_q)$.

Local indices. Recenter an axial equilibrium at the origin. Axisymmetry and three-dimensional harmonicity give the convergent zonal expansion

$$V(r,\theta) - V(0) = \sum_{j\geq 1} \mathsf{c}_j r^j P_j(\cos\theta),$$

convergent, together with its term-by-term derivatives, uniformly on compact subsets of a ball about the origin containing no site. Criticality gives $\mathsf{c}_1 = 0$. The potential cannot be locally constant: otherwise real-analytic continuation on the connected complement of the sites would make it constant, contradicting its pole at a site. Let $k = \min\{j : \mathsf{c}_j \neq 0\} \geq 2$. For $\mathcal{Y}_k = r^k P_k(\cos\theta)$ the meridian gradient on the unit circle is

$$\nabla \mathcal{Y}_k(\theta) = e^{i\theta} g_k(\theta), \qquad g_k = kP_k(\cos\theta) - i\sin\theta\, P_k'(\cos\theta).$$

The function g_k never vanishes: a simultaneous zero of its real and imaginary parts with $\sin\theta \neq 0$ would be a common zero of P_k and P_k', whereas the roots of P_k are simple [12, Theorem 3.3.1], and at $\sin\theta = 0$ one has $P_k(\pm 1) \neq 0$. At every zero of $P_k(\cos\theta)$ we have $\Re g_k = 0$ and $(\Re g_k)' = -k\sin\theta P_k'(\cos\theta) = k\,\Im g_k$, so

$$\frac{d}{d\theta}\arg g_k = \frac{\Re g_k\,(\Im g_k)' - \Im g_k\,(\Re g_k)'}{|g_k|^2} = -k.$$

There are $2k$ such crossings on $0 \leq \theta < 2\pi$, all clockwise. They are the transverse preimages of the vertical projective direction and compute the degree of the projectivized direction map as $-2k$; since that degree is twice the ordinary winding, $\mathrm{wind}(g_k) = -k$ and

$$\mathrm{ind}_0\, \nabla_{P_{\mathrm{m}}} V = 1 - k \leq -1.$$

The higher terms do not change the index: after division by r^{k-1} their gradients converge uniformly to zero on the unit circle, while the leading field has a positive minimum norm.

Winding at infinity. Put $M_\ell = \sum_i q_i h_i^\ell$. For $R > \max_i |h_i|$ the Legendre generating function gives the uniformly differentiable expansion $V(R,\theta) = \sum_{\ell\geq 0} M_\ell R^{-\ell-1} P_\ell(\cos\theta)$. The Vandermonde matrix $(h_i^\ell)_{0\leq\ell\leq 2,\,1\leq i\leq 3}$ is invertible, and the charge vector is nonzero, so the first nonvanishing moment has order $\ell \in \{0,1,2\}$. The leading meridian gradient is a nonzero scalar multiple of $e^{i\theta}\widetilde{g}_\ell(\theta)$ with

$$\widetilde{g}_\ell = -(\ell+1)P_\ell(\cos\theta) - i\sin\theta\, P_\ell'(\cos\theta),$$

which never vanishes by the same argument as for g_k (and equals -1 when $\ell = 0$). At a root of $P_\ell(\cos\theta)$ one has $(\Re\widetilde{g}_\ell)' = -(\ell+1)\Im\widetilde{g}_\ell$, so $\frac{d}{d\theta}\arg\widetilde{g}_\ell = \ell + 1 > 0$; the 2ℓ crossings are all counterclockwise, whence $\mathrm{wind}(\widetilde{g}_\ell) = \ell$ and

$$w_\infty = \ell + 1.$$

The uniform remainder cannot alter the winding on a sufficiently large circle.

Proof of Theorem 1, collinear case. Apply the degree theorem in P_{m}. The finite critical set gives finitely many isolated zeros of $\nabla_{P_{\mathrm{m}}} V_q$, and every site again contributes $+1$, so with $\mathrm{Crit}(V_q) = \{x_1, \ldots, x_{\mathfrak{N}}\}$

$$\sum_{j=1}^{\mathfrak{N}} \mathrm{ind}_{x_j}\, \nabla_{P_{\mathrm{m}}} V_q = w_\infty - 3 = \ell - 2.$$

Every summand is at most -1, so $\mathfrak{N} \leq 2 - \ell \leq 2$. The collinear case is therefore stronger than required, and the proof of Theorem 1 is complete. □

10 Computer-assisted exact verification

Computer algebra verifies (3.2)–(3.4), (3.8), (3.9), (3.10) and (3.11); the equilateral eliminant, critical values and Hessian data in the proof of Proposition 3.4; (5.5), (5.6), the affine-compatibility reduction in Step 2 of Proposition 5.1, (5.9)–(5.10), and the chart-boundary limits (5.11)–(5.12); the sourcewise derivative table of Section 7; and the kernel, Laurent, and positivity identities leading to (7.3) and Lemma 7.2. Proposition 3.3, Proposition 5.1, and all analytic, topological and degree-theoretic implications are proved in the text.

The directory `certificates/` in the github repository `github.com/shivakintali/Maxwell3` contains the exact certificates and a README file relating each script to the displayed identities it checks. From the manuscript directory, a clean environment reproducing the audited run is created by

```
python3.13 -m venv .venv
.venv/bin/python -m pip install -r certificates/requirements.txt
.venv/bin/python -I certificates/run_all.py
```

Every proof-critical comparison is enforced by a Python assertion. The runner requires CPython 3.13, rejects optimized mode (which would disable assertions) and non-isolated mode, and pins the SymPy and mpmath versions. The scripts reconstruct their expressions from the displayed definitions. The canonical suite uses exact symbolic differentiation and simplification over integer, rational, algebraic-number, polynomial, rational-function and Laurent-polynomial expressions, including exact resultants and limits. It performs no floating-point computation and loads no saved symbolic objects. The suite was audited with CPython 3.13.9, SymPy 1.14.0 and mpmath 1.3.0; the exact runtime is recorded in `certificates/runtime.txt`. After installing the separately pinned dependencies from `certificates/requirements-sanity.txt`, the optional command

```
.venv/bin/python -I certificates/interior_jacobian.py --sanity
```

adds a floating-point consistency check of (3.3), which is not used anywhere in the proof.

A The interior Jacobian certificate

Table 1 lists the thirty AM–GM circuits of (3.4). Each row records the weight n_r and the exponent vectors $\alpha_r, \gamma_r, \beta_r \in \mathbb{Z}^6_{\geq 0}$ in the variables $X = (\lambda_1, \lambda_2, \lambda_3, y_1, y_2, y_3)$; in every row $\alpha_r + \gamma_r = 2\beta_r$, the monomials X^{α_r} and X^{γ_r} have positive coefficients in $\mathcal{V}_{\mathrm{Ravi}}$, and the thirty β_r exhaust the monomials with negative coefficients. Subtracting $\sum_r \frac{\mathsf{n}_r}{2}(X^{\alpha_r} + X^{\gamma_r} - 2X^{\beta_r})$ from $\mathcal{V}_{\mathrm{Ravi}}$ leaves the tail $\mathcal{T}$, which has 1548 terms with positive integer coefficients, the smallest being 16.

Table 1: The thirty AM–GM circuits of (3.4).

n_r	α_r						γ_r						β_r					
448	5	5	1	0	6	0	7	1	3	2	4	0	6	3	2	1	5	0
448	5	1	5	0	0	6	7	3	1	2	0	4	6	2	3	1	0	5
448	4	5	2	0	6	0	6	1	4	2	4	0	5	3	3	1	5	0
448	4	2	5	0	0	6	6	4	1	2	0	4	5	3	3	1	0	5
448	1	7	3	4	2	0	5	5	1	6	0	0	3	6	2	5	1	0
448	1	6	4	4	2	0	5	4	2	6	0	0	3	5	3	5	1	0
448	2	4	5	0	0	6	4	6	1	0	2	4	3	5	3	0	1	5

n_r	α_r						γ_r						β_r					
448	1	4	6	4	0	2	5	2	4	6	0	0	3	3	5	5	0	1
448	2	5	4	0	6	0	4	1	6	0	4	2	3	3	5	0	5	1
448	1	3	7	4	0	2	5	1	5	6	0	0	3	2	6	5	0	1
448	1	5	5	0	0	6	3	7	1	0	2	4	2	6	3	0	1	5
448	1	5	5	0	6	0	3	1	7	0	4	2	2	3	6	0	5	1
192	6	4	1	0	6	0	8	2	1	2	4	0	7	3	1	1	5	0
192	6	1	4	0	0	6	8	1	2	2	0	4	7	1	3	1	0	5
192	2	8	1	4	2	0	4	6	1	6	0	0	3	7	1	5	1	0
192	2	1	8	4	0	2	4	1	6	6	0	0	3	1	7	5	0	1
192	1	6	4	0	0	6	1	8	2	0	2	4	1	7	3	0	1	5
192	1	2	8	0	4	2	1	4	6	0	6	0	1	3	7	0	5	1
32	7	4	0	0	6	0	9	2	0	2	4	0	8	3	0	1	5	0
32	7	0	4	0	0	6	9	0	2	2	0	4	8	0	3	1	0	5
32	6	5	0	0	6	0	8	3	0	2	4	0	7	4	0	1	5	0
32	6	0	5	0	0	6	8	0	3	2	0	4	7	0	4	1	0	5
32	3	8	0	4	2	0	5	6	0	6	0	0	4	7	0	5	1	0
32	3	0	8	4	0	2	5	0	6	6	0	0	4	0	7	5	0	1
32	2	9	0	4	2	0	4	7	0	6	0	0	3	8	0	5	1	0
32	2	0	9	4	0	2	4	0	7	6	0	0	3	0	8	5	0	1
32	0	7	4	0	0	6	0	9	2	0	2	4	0	8	3	0	1	5
32	0	6	5	0	0	6	0	8	3	0	2	4	0	7	4	0	1	5
32	0	3	8	0	4	2	0	5	6	0	6	0	0	4	7	0	5	1
32	0	2	9	0	4	2	0	4	7	0	6	0	0	3	8	0	5	1

B The edge-fold polynomial $\mathcal{S}$

The polynomial $\mathcal{S}$ defined by the exact quotient (3.10) is $\sum_{j=0}^{6} \mathcal{S}_j s^j$ with the following coefficients, each homogeneous of degree 8 in (ℓ_1, ℓ_2, ℓ_3), so that $\mathcal{S}$ has total degree 14. As a whole, $\mathcal{S}$ is invariant under the reflection $(\ell_2, \ell_3, s) \mapsto (\ell_3, \ell_2, 1-s)$ of the edge chamber, which permutes the coefficients $\mathcal{S}_j$ among themselves.

$$
\begin{aligned}
\mathcal{S}_6 =\; & (\ell_1 - 3\ell_2 - 3\ell_3)^2 (\ell_1 + 3\ell_2 + 3\ell_3)^2 \left(16\ell_1^2\ell_2\ell_3 - 3\ell_2^4 + 6\ell_2^2\ell_3^2 - 3\ell_3^4\right), \\
\mathcal{S}_5 =\; & -(\ell_1 - 3\ell_2 - 3\ell_3)(\ell_1 + 3\ell_2 + 3\ell_3)\big(48\ell_1^4\ell_2\ell_3 - 13\ell_1^2\ell_2^4 - 612\ell_1^2\ell_2^3\ell_3 - 846\ell_1^2\ell_2^2\ell_3^2 \\
& - 252\ell_1^2\ell_2\ell_3^3 - 5\ell_1^2\ell_3^4 + 157\ell_2^6 + 126\ell_2^5\ell_3 - 381\ell_2^4\ell_3^2 - 324\ell_2^3\ell_3^3 \\
& + 219\ell_2^2\ell_3^4 + 198\ell_2\ell_3^5 + 5\ell_3^6\big), \\
\mathcal{S}_4 =\; & 2\big(26\ell_1^6\ell_2\ell_3 - 11\ell_1^4\ell_2^4 - 723\ell_1^4\ell_2^3\ell_3 - 926\ell_1^4\ell_2^2\ell_3^2 - 273\ell_1^4\ell_2\ell_3^3 - \ell_1^4\ell_3^4 + 282\ell_1^2\ell_2^6 \\
& + 4864\ell_1^2\ell_2^5\ell_3 + 12542\ell_1^2\ell_2^4\ell_3^2 + 12340\ell_1^2\ell_2^3\ell_3^3 + 5102\ell_1^2\ell_2^2\ell_3^4 + 724\ell_1^2\ell_2\ell_3^5 \\
& + 2\ell_1^2\ell_3^6 - 1711\ell_2^8 - 3087\ell_2^7\ell_3 + 3352\ell_2^6\ell_3^2 + 9585\ell_2^5\ell_3^3 + 2964\ell_2^4\ell_3^4 \\
& - 4725\ell_2^3\ell_3^5 - 3308\ell_2^2\ell_3^6 - 477\ell_2\ell_3^7 - \ell_3^8\big), \\
\mathcal{S}_3 =\; & -2\ell_2\big(12\ell_1^6\ell_3 - 9\ell_1^4\ell_2^3 - 471\ell_1^4\ell_2^2\ell_3 - 427\ell_1^4\ell_2\ell_3^2 - 81\ell_1^4\ell_3^3 + 298\ell_1^2\ell_2^5 \\
& + 4030\ell_1^2\ell_2^4\ell_3 + 8036\ell_1^2\ell_2^3\ell_3^2 + 5780\ell_1^2\ell_2^2\ell_3^3 + 1602\ell_1^2\ell_2\ell_3^4 + 126\ell_1^2\ell_3^5 \\
& - 2209\ell_2^7 - 2211\ell_2^6\ell_3 + 6123\ell_2^5\ell_3^2 + 8541\ell_2^4\ell_3^3 - 147\ell_2^3\ell_3^4 - 3681\ell_2^2\ell_3^5 \\
& - 1175\ell_2\ell_3^6 - 57\ell_3^7\big), \\
\mathcal{S}_2 =\; & \ell_2\big(4\ell_1^6\ell_3 - 7\ell_1^4\ell_2^3 - 282\ell_1^4\ell_2^2\ell_3 - 142\ell_1^4\ell_2\ell_3^2 - 12\ell_1^4\ell_3^3 + 334\ell_1^2\ell_2^5 \\
& + 3444\ell_1^2\ell_2^4\ell_3 + 4706\ell_1^2\ell_2^3\ell_3^2 + 2060\ell_1^2\ell_2^2\ell_3^3 + 284\ell_1^2\ell_2\ell_3^4 + 12\ell_1^2\ell_3^5
\end{aligned}
$$

$$
\begin{aligned}
&- 3207\ell_2^7 - 1246\ell_2^6\ell_3 + 8432\ell_2^5\ell_3^2 + 6268\ell_2^4\ell_3^3 - 1843\ell_2^3\ell_3^4 - 1778\ell_2^2\ell_3^5 \\
&- 142\ell_2\ell_3^6 - 4\ell_3^7\big),\\
\mathcal{S}_1 &= \ell_2^3\big(\ell_1^4\ell_2 + 30\ell_1^4\ell_3 - 90\ell_1^2\ell_2^3 - 704\ell_1^2\ell_2^2\ell_3 - 482\ell_1^2\ell_2\ell_3^2 - 60\ell_1^2\ell_3^3 \\
&\quad + 1241\ell_2^5 - 46\ell_2^4\ell_3 - 2370\ell_2^3\ell_3^2 - 632\ell_2^2\ell_3^3 + 481\ell_2\ell_3^4 + 30\ell_3^5\big),\\
\mathcal{S}_0 &= 8\ell_2^5\big(\ell_1^2\ell_2 + 7\ell_1^2\ell_3 - 25\ell_2^3 + 7\ell_2^2\ell_3 + 25\ell_2\ell_3^2 - 7\ell_3^3\big).
\end{aligned}
$$

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